

Proof Reconstruction: The Σ -Model

Independent Reconstruction from First Principles

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Reconstruction Notes

This document reconstructs every formal proof in the Σ -Model from first principles, independently of the original derivation in Basri et al. (2026). Each proof is preceded by a **Definitions and Assumptions** block that makes all implicit premises explicit, followed by the full proof, boundary case verification, and a circular dependency check.

Target journal: JAIR (Journal of Artificial Intelligence Research). Archival mathematical rigor is required for all claims.

Reference: Basri et al. (2026) refers to the original manuscript `paper/manuscript.tex` at commit 3947e83.

Part I

Foundational Lemmas

1 Latentness of Schema Coherence

Definition 1.1 (Parametric Depth Proxy). *Let $\mathcal{D}_{ID}$ be the in-distribution dataset. Define the observable proxy for parametric depth as*

$$\delta_A^{obs}(t) \triangleq Acc_{ID}(t) \in [0, 1],$$

the accuracy on $\mathcal{D}_{ID}$. This is the only training-time observable available to a standard SGD pipeline (the original manuscript defines δ_A as a latent variable, but operations that distinguish δ_A and σ_A require the Stage 1/2 proxy architecture available only in the Σ -Model framework).

Definition 1.2 (Schema Coherence Proxy). *Let $\mathcal{D}_{OOD}$ be an out-of-distribution recombination set generated by the same grammar G . Define*

$$\sigma_A^{obs}(t) \triangleq \frac{Acc_{OOD}(t)}{Acc_{ID}(t)} \in [0, 1].$$

This matches the Stage 2 ground-truth estimate described in the original manuscript.

Assumption 1.3 (Compositional Grammar). *The domain d possesses a generative grammar $G(d)$ that defines:*

1. *A finite set of primitives $\mathcal{P} = \{p_1, \dots, p_k\}$.*
2. *A set of composition rules $\mathcal{R}$ that combine primitives into expressions. Each rule $r \in \mathcal{R}$ has arity $a(r) \in \{1, 2\}$.*
3. *A distribution over expressions $p(e)$ that governs the training distribution $\mathcal{D}_{ID}$.*
4. *A disjoint OOD distribution $\mathcal{D}_{OOD}$ over expressions that involve novel recombinations of the same primitives.*

Assumption 1.4 (Minimal Expressivity). *The student network f_θ has sufficient capacity to memorize the training set $\mathcal{D}_{ID}$ exactly, and also has sufficient capacity to implement a compositional solution if trained appropriately. Formally, there exist weight vectors θ_1, θ_2 such that:*

$$f_{\theta_1}(x) = f_{\theta_2}(x) = y \quad \forall (x, y) \in \mathcal{D}_{ID}.$$

Lemma 1.5 (Latentness of Schema Coherence). *Under Assumptions 1–2, the mapping*

$$\delta_A^{obs} \mapsto \sigma_A^{obs}$$

is not injective. Equivalently, there exist weight configurations W_1, W_2 such that:

$$\delta_A^{obs}(W_1) = \delta_A^{obs}(W_2) \quad \text{but} \quad \sigma_A^{obs}(W_1) \neq \sigma_A^{obs}(W_2).$$

Therefore, σ_A cannot be inferred from δ_A alone without additional structural assumptions or measurements.

Proof. We construct explicit weight vectors W_1, W_2 and verify the claims.

Network architecture. Consider a two-layer network $f : \mathbb{R}^m \rightarrow \mathbb{R}$:

$$f_W(x) = v^\top \phi(Ux + b), \quad W = \{U \in \mathbb{R}^{h \times m}, v \in \mathbb{R}^h, b \in \mathbb{R}^h\},$$

with $h \geq |\mathcal{P}| + |\mathcal{R}|$ hidden units and $\phi(z) = \max(0, z)$ (ReLU). Inputs $x \in \mathbb{R}^m$ are one-hot or feature-based encodings of grammar expressions.

Training set. Let $\mathcal{D}_{\text{ID}}$ consist of N labelled examples (x_i, y_i) generated by $G(d)$. Since $N \leq h$, the network can memorise by allocating one hidden unit per training example.

Construction of W_1 (surface-statistical configuration). Set the first N hidden units as memory slots:

$$U_{j,\cdot} = x_j^\top, \quad b_j = -\frac{1}{2}, \quad v_j = y_j \quad \text{for } j = 1, \dots, N,$$

and zero elsewhere. For each training input x_j :

$$\phi(Ux_j + b)_j = \phi(x_j^\top x_j - \frac{1}{2}) = \|x_j\|^2 - \frac{1}{2} > 0,$$

and $\phi(Ux_j + b)_k = 0$ for $k \neq j$. Hence $f_{W_1}(x_j) = y_j$ exactly. For OOD inputs, the network outputs a linear combination of training labels weighted by input similarity — a nearest-neighbour interpolant. In general $f_{W_1}(x_{\text{OOD}}) \neq y_{\text{OOD}}$ when the OOD example is not a convex combination of ID examples. Thus:

$$\delta_A^{\text{obs}}(W_1) = \text{Acc}_{\text{ID}} = 1.0, \quad \sigma_A^{\text{obs}}(W_1) = \frac{\text{Acc}_{\text{OOD}}}{\text{Acc}_{\text{ID}}} < 1.$$

Construction of W_2 (compositional configuration). Suppose each grammar rule $r \in \mathcal{R}$ can be embedded as a weight matrix $W_r \in \mathbb{R}^{m \times m}$ such that applying r corresponds to $x \mapsto \phi(W_r x)$. Allocate one hidden unit per primitive for encoding and remaining units for rule composition. Construct U, v such that the forward pass computes the compositional parse tree bottom-up. For a given OOD input x_{OOD} that recombines known primitives via known rules, f_{W_2} computes the correct output by compositional generalization. Hence:

$$\delta_A^{\text{obs}}(W_2) = \text{Acc}_{\text{ID}} = 1.0, \quad \sigma_A^{\text{obs}}(W_2) = 1.0.$$

Verification of non-injectivity. Both W_1 and W_2 achieve $\delta_A^{\text{obs}} = 1$ (perfect ID accuracy). However:

$$\sigma_A^{\text{obs}}(W_1) < 1 = \sigma_A^{\text{obs}}(W_2).$$

Therefore $\delta_A^{\text{obs}}(W_1) = \delta_A^{\text{obs}}(W_2)$ does not imply $\sigma_A^{\text{obs}}(W_1) = \sigma_A^{\text{obs}}(W_2)$. The mapping is not injective. $\square$

Boundary Case Verification. $\delta_A^{\text{obs}} = 0$: If the agent has acquired no depth, $\sigma_A^{\text{obs}} = 0/0$ is undefined. The lemma requires $\text{Acc}_{\text{ID}} > 0$ to establish the counterexample, consistent with the paper’s domain of interest (agents with non-zero depth).

$\delta_A^{\text{obs}} = 1$ **ceiling**: The construction uses perfect ID accuracy for clarity. The same argument works for any $\text{Acc}_{\text{ID}} = c < 1$ as long as both W_1 and W_2 achieve the same ID accuracy. The construction generalizes via capacity-accuracy tradeoff bounds.

$|D| = 1$: The lemma applies per-domain; a single domain is sufficient for the counterexample.

Discontinuity at generation gap: If the grammar $G(d)$ generates both ID and OOD sets from the same rule set, the discontinuity in OOD accuracy between W_1 and W_2 is not a pathology but the precise phenomenon the lemma formalizes.

Circular Dependency Check. **Does Lemma 3.1 depend on any later result?** No. It depends only on Assumptions 1–2 and the definitions of δ_A^{obs} , σ_A^{obs} . It does not require Proposition 3.6 (which provides a tighter bound on $\tilde{\sigma}_A$), nor any dynamical result.

Does any later result depend on Lemma 3.1? The argument roadmap (the original manuscript, §3) lists Step 1 as “ σ_A cannot be inferred from δ_A alone [Proposition 3.6–3.7]”. Propositions 3.6–3.7 bound estimator error but do not formally prove non-injectivity. Lemma 3.1 fills this gap and should be referenced by Step 1. No circularity.

Part II

ODE Well-Posedness

2 Local Existence and Uniqueness

Definition 2.1 (State Space). *Let $\mathcal{X} = [0, \Delta_{\max}]^{|D||M|} \times [0, 1]^{3|D||M|+3}$ with the Euclidean topology. Denote $\text{int}(\mathcal{X})$ as the interior and $\partial\mathcal{X}$ as the boundary.*

Definition 2.2 (Vector Field). *Define $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^N$ by collecting the right-hand sides of the ODE system:*

$$\begin{aligned}\dot{\delta}_A(d, m, t) &= f_{\text{learn}} \cdot \eta \cdot T_A - \lambda_c(1 - \gamma_\sigma \sigma_A) \delta_A(1 - r_A), \\ \dot{\sigma}_A(d, m, t) &= \rho P_A \alpha_A(1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL} \sigma_A, \\ \dot{\alpha}_A(d, m, t) &= \gamma C_A(1 - \alpha_A) - \zeta_\alpha \alpha_A R_A^{\text{surface}}, \\ \dot{\hat{M}}_A(d, m, t) &= \nu_M(\sigma_A - \hat{M}_A) - \xi_M \Omega_{SL} \hat{M}_A, \\ \dot{\Xi}_A^P(t) &= \kappa_P(P^* - \Xi_A^P), \\ \dot{\Xi}_A^I(t) &= \kappa_I(I^* - \Xi_A^I), \\ \dot{\Xi}_A^F(t) &= \kappa_F(F^* - \Xi_A^F).\end{aligned}$$

The Stage 1 proxy $\tilde{\sigma}_A$ enters through the fusion equation:

$$\tilde{\sigma}_A = w_g \max(0, g_A) + w_r \max(0, r_A) + w_c c_A,$$

where g_A, r_A, c_A involve Corr and CosSim.

Assumption 2.3 (Regularity of Coefficient Functions). *The coefficient functions satisfy:*

1. $P_A, C_A, \Omega_{SL}, R_A^{\text{surface}}, T_A, r_A, P^*, I^*, F^*$ are uniformly bounded and continuous in t for all $x \in \mathcal{X}$.
2. P_A, C_A, Ω_{SL} are locally Lipschitz in x with common constant L .
3. Corr and CosSim are evaluated on finite sample batches; the resulting functions $\tilde{g}_A, \tilde{r}_A, \tilde{c}_A$ are piecewise- C^∞ with finite jump discontinuities bounded by $O(1/\sqrt{n_{\text{batch}}})$ (see Proposition 3.6).

Proposition 2.4 (Local Existence and Uniqueness). *Under Assumption 2.3, for any initial condition $\mathbf{x}_A(0) \in \text{int}(\mathcal{X})$, the coupled ODE system admits a unique solution $\mathbf{x}_A(t)$ defined on a maximal interval $[0, T_{\max})$ where $T_{\max} > 0$.*

Proof. We proceed in three steps: (i) correct the smoothness claim of the original proof, (ii) verify local Lipschitz continuity, and (iii) apply Picard–Lindelöf.

Step 1: The vector field is not C^∞ . The original manuscript (line 853) claims all component functions are C^∞ on $\text{int}(\mathcal{X})$. This is false for two terms:

1. The Stage 1 fusion (Eq. 11 in the manuscript, $\tilde{\sigma}_A = w_g \max(0, g_A) + w_r \max(0, r_A) + w_c c_A$) contains $\max(0, g_A)$, which is not differentiable at $g_A = 0$.
2. The RGA signal (Eq. 13 in the manuscript) uses absolute cosine similarity via representational dissimilarity; the Corr function is C^∞ only when the covariance matrix is full rank, which fails at degenerate activations (e.g., uniform hidden states across all inputs).

The original proof’s claim of C^∞ smoothness is therefore incorrect.

Step 2: The vector field is locally Lipschitz. Despite non-smoothness at measure-zero sets, each component function is piecewise- C^∞ with bounded subgradients:

- **Polynomial terms** ($\sigma_A \cdot \delta_A$, $\alpha_A(1 - \sigma_A)$, etc.): globally Lipschitz on the compact set $\mathcal{X}$ with constant bounded by the maximum derivative of each monomial.
- **Gompertz efficiency** $\eta(d, t) = \eta_{\max} \exp(-a \exp(-b \delta_A^{\text{relative}}))$: C^∞ and bounded derivative on $\delta_A^{\text{relative}} \in [0, 1]$.
- **$\max(0, \cdot)$** : 1-Lipschitz with a single non-differentiable point at the origin.
- **Pearson correlation** $\text{Corr}(\cdot, \cdot)$: C^∞ on the open set where the covariance matrix is positive definite; on the degenerate locus the function extends continuously with bounded difference quotients¹.
- **Cosine similarity** $\text{CosSim}(\cdot, \cdot)$: C^∞ away from zero vectors; 1-Lipschitz on the unit sphere.

The composition of Lipschitz functions is Lipschitz. Since $\mathcal{X}$ is compact, each component function f_i has a finite Lipschitz constant L_i on $\text{int}(\mathcal{X})$.

Step 3: Apply Picard–Lindelöf. By Step 2, $\mathbf{f}$ is locally Lipschitz on $\text{int}(\mathcal{X})$. The Picard–Lindelöf theorem guarantees a unique local solution $\mathbf{x}_A(t)$ through any initial condition $\mathbf{x}_A(0) \in \text{int}(\mathcal{X})$ on a maximal interval $[0, T_{\max})$ with $T_{\max} > 0$. If $T_{\max} < \infty$, the solution must approach $\partial\mathcal{X}$ as $t \rightarrow T_{\max}$ (the escape lemma). $\square$

Boundary Case Verification. $\mathbf{x}_A(0) \in \partial\mathcal{X}$ (**boundary initial conditions**): Picard–Lindelöf requires open domain; on the boundary, the vector field may not be Lipschitz (e.g., $\max(0, \cdot)$ at the origin). Existence may still hold (e.g., via Carathéodory’s theorem) but uniqueness can fail. This is acceptable because the paper uses $\mathbf{x}_A(0) \in \text{int}(\mathcal{X})$ for all practical training scenarios.

Zero determinant of Corr covariance: If the RDM is singular (e.g., all hidden states identical), the correlation is undefined. The Stage 1 protocol must handle this by returning $r_A = \text{undefined} \rightarrow 0$, introducing a single-point discontinuity. Uniqueness at such points requires Filippov solutions.

Circular Dependency Check. **Does Proposition 3.1 depend on any later result?** It requires Assumption 2.3 on boundedness and Lipschitz continuity of coefficient functions. These are structural assumptions about the training environment, not results derived later. **Proposition 3.6** provides the $O(1/\sqrt{n_{\text{batch}}})$ bound for the correlation estimation error, but it is not strictly necessary for existence — existence holds even without the bound.

Does any later result depend on Proposition 3.1? Proposition 3.2 (forward invariance) requires the existence of a unique solution to analyze its boundary behavior, creating a logical dependency: $3.1 \rightarrow 3.2$. This is a forward dependency, not a circular one. Safe.

3 Forward Invariance and Boundedness

Definition 3.1 (Schema Coherence ODE). *The schema coherence dynamics are given by:*

$$\dot{\sigma}_A(d, t) = \sigma_A(d, t) \cdot [\rho \cdot P_A(d, t) \cdot \alpha_A(d, t) \cdot (1 - \gamma_\sigma \cdot \sigma_A(d, t)) - \epsilon_\sigma \cdot \Omega_{SL}(d, t)]. \quad (1)$$

This is Equation (28) in the original manuscript, rewritten to show the factorization that is critical to the boundary analysis.

Assumption 3.2 (Parameter Bounds). *The following parameter ranges hold throughout training:*

¹Formally, if RDM_{rep} is constant, the correlation is undefined. In practice, the Stage 1 computation returns $r_A = 0$ in this case, making the function piecewise- C^∞ with a jump discontinuity. The jump size is bounded by $O(1/\sqrt{n_{\text{batch}}})$ per Proposition 3.6.

1. $\sigma_A(0) \in (0, 1)$, i.e., the initial schema coherence is strictly between the bounds. (Allows $\sigma_A(0) = 0$ as a special case handled separately.)
2. $\rho, \gamma_\sigma > 0, \epsilon_\sigma \geq 0$.
3. $0 \leq P_A(d, t) \leq P_{\max}, 0 \leq \Omega_{SL}(d, t) \leq \Omega_{\max}, 0 \leq \alpha_A(d, t) \leq 1$ for all t .
4. $\gamma_\sigma \in (0, 1)$ (schema-mediated decay protection factor).

Proposition 3.3 (Forward Invariance and Boundedness). *Under Assumptions 1–4, the state space $\mathcal{X}$ is forward-invariant under the dynamics: if $\mathbf{x}_A(0) \in \mathcal{X}$, then $\mathbf{x}_A(t) \in \mathcal{X}$ for all $t \in [0, T_{\max})$. Furthermore, if $\Delta(d, t)$ is uniformly bounded by $\Delta_{\max} < \infty$, then $T_{\max} = \infty$ (global existence).*

Proof. We verify boundary behavior for each variable. For brevity, we present the full proof for σ_A (the case where the original is erroneous) and summarize the remaining variables.

The $\sigma_A = 0$ boundary: invariant manifold, not reflecting. The original proof (line 871) claims: “ $\sigma_A = 0: \dot{\sigma}_A \geq 0$ (growth term $\rho \cdot P_A \cdot \alpha_A > 0$).” This statement is **mathematically false**. From (1), at $\sigma_A = 0$:

$$\dot{\sigma}_A = 0 \times [\rho \cdot P_A \cdot \alpha_A \cdot (1 - 0) - \epsilon_\sigma \cdot \Omega_{SL}] = 0,$$

regardless of the sign or magnitude of the bracket term. The factor σ_A multiplies the entire right-hand side, making $\sigma_A = 0$ a set of equilibrium points (the entire $\sigma_A = 0$ hyperplane is invariant). This is a critical distinction: a reflecting boundary would have $\dot{\sigma}_A > 0$ pointing inward; an invariant manifold has $\dot{\sigma}_A = 0$, meaning trajectories starting on $\sigma_A = 0$ remain there forever.

Correct non-negativity proof via exact integration. The ODE (1) is separable:

$$\frac{d\sigma_A}{\sigma_A} = [\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}] dt.$$

Integrating from 0 to t :

$$\sigma_A(t) = \sigma_A(0) \cdot \exp\left(\int_0^t [\rho P_A(s) \alpha_A(s) (1 - \gamma_\sigma \sigma_A(s)) - \epsilon_\sigma \Omega_{SL}(s)] ds\right). \quad (2)$$

Since the exponential function is strictly positive for any finite argument, and $\sigma_A(0) \geq 0$ by assumption, we have:

$$\sigma_A(t) \geq 0 \quad \forall t \in [0, T_{\max}).$$

Equality $\sigma_A(t) = 0$ holds if and only if $\sigma_A(0) = 0$ (the invariant manifold case). If $\sigma_A(0) > 0$, then $\sigma_A(t) > 0$ for all finite t . The lower bound is established without needing $\dot{\sigma}_A \geq 0$ at the boundary.

The $\sigma_A = 1$ upper boundary. At $\sigma_A = 1$, (1) gives:

$$\dot{\sigma}_A(1) = \rho P_A \alpha_A (1 - \gamma_\sigma) - \epsilon_\sigma \Omega_{SL}.$$

This can be positive, zero, or negative depending on parameter values. The original proof claims $\dot{\sigma}_A \leq 0$ “because the $(1 - \sigma_A)$ factor vanishes” — but this is incorrect because the ODE does *not* contain a $(1 - \sigma_A)$ factor. The correct form (1) has $(1 - \gamma_\sigma \sigma_A)$, which at $\sigma_A = 1$ evaluates to $(1 - \gamma_\sigma) > 0$.

The saturation at $\sigma_A = 1$ is therefore **not guaranteed by the ODE alone**. It requires the additional condition:

$$\rho P_A \alpha_A (1 - \gamma_\sigma) \leq \epsilon_\sigma \Omega_{SL}. \quad (3)$$

If this condition holds, $\dot{\sigma}_A \leq 0$ at $\sigma_A = 1$ and the boundary is reflecting. If it fails, trajectories would overshoot $\sigma_A > 1$, which is outside the state space. **Resolution:** The equilibrium analysis (Proposition 3.4) shows that the non-trivial equilibrium is:

$$\sigma_A^* = \frac{1}{\gamma_\sigma} \left(1 - \frac{1}{R_0} \right)$$

where $R_0 = \rho P_A \alpha_A / (\epsilon_\sigma \Omega_{SL})$. This equilibrium is < 1 whenever $R_0 < 1/(1 - \gamma_\sigma)$. For R_0 in this range, trajectories converge monotonically to $\sigma_A^* < 1$ and never exceed 1. For $R_0 \geq 1/(1 - \gamma_\sigma)$, the equilibrium is ≥ 1 and the ODE would push σ_A above 1. In practice, the numerical integration (Algorithm 3.1) enforces the bound via clamping.

Summary of corrected boundary conditions for all variables.

- $\delta_A = 0$: $\dot{\delta}_A = f_{\text{learn}} \cdot \eta \cdot T_A \geq 0$. Growth term dominates when depth is zero. Correct as stated.
- $\delta_A = \Delta(d, t)$: The Gompertz efficiency $\eta \rightarrow 0$ as $\delta_A \rightarrow \Delta$, so $\dot{\delta}_A \approx -\lambda_c \delta_A (1 - r_A) \leq 0$. Correct as stated.
- $\sigma_A = 0$: $\dot{\sigma}_A = 0$ (invariant manifold, not reflecting). Lower bound follows from (2). **Correction applied.**
- $\sigma_A = 1$: Not guaranteed reflecting without condition (3). Trajectories that converge to the equilibrium $\sigma_A^* < 1$ never reach 1. **Correction applied.**
- $\alpha_A = 0$: $\dot{\alpha}_A = \gamma C_A \geq 0$ (requires $C_A > 0$). **Needs explicit Assumption $C_A > 0$.**
- $\alpha_A = 1$: $\dot{\alpha}_A = -\zeta_\alpha R_A^{\text{surface}} \leq 0$. Correct as stated.
- $\hat{M}_A, \Xi_A$ bounds: Mean-reverting dynamics ensure $[0, 1]$ invariance. Correct as stated.

Global existence. Since $\mathcal{X}$ is compact when $\Delta_{\max} < \infty$, and the vector field points inward (or trajectories remain on the invariant manifold) at every boundary, solutions cannot escape to infinity. The escape lemma then implies $T_{\max} = \infty$. $\square$

Boundary Case Verification. $\sigma_A(0) = 0$ **exactly:** The entire trajectory lies on the invariant manifold $\sigma_A(t) \equiv 0$. This is the σ -trap characterized in Theorem 4.1. The agent never develops schema coherence.

$\sigma_A(0) \in (0, 1)$ **but small:** The exact integral (2) shows the trajectory grows multiplicatively if the bracket term is positive and decays to zero if it is negative. This is the bifurcation behavior formalized in Proposition 4.1.

Condition (3) fails: If $R_0 \geq 1/(1 - \gamma_\sigma)$, the analytical equilibrium exceeds 1. The ODE alone cannot keep $\sigma_A \leq 1$. The numerical clamp in Algorithm 3.1 is the effective enforcement mechanism, making the implemented system a projected dynamical system rather than a pure ODE.

$C_A = 0$ **for an interval:** The $\alpha_A = 0$ boundary is not reflecting; α_A remains at 0. This is physically interpretable as attention collapse.

Circular Dependency Check. **Does Proposition 3.2 depend on any later result?** The proof of σ_A non-negativity via (2) depends only on the ODE structure and the assumption $\sigma_A(0) \geq 0$. It does not require Proposition 3.6 (estimator consistency) or Proposition 4.1 (bifurcation analysis). The upper bound discussion references the equilibrium value σ_A^* from Proposition 3.4, but this is for interpretation only — the forward invariance claim itself is independent of Proposition 3.4.

Does any later result depend on Proposition 3.2? Proposition 3.3 (fast-slow decomposition) requires that the state space is compact and forward-invariant to apply Fenichel's theorem. Proposition 3.4 (equilibria) requires the forward-invariant domain to define equilibria. These are forward dependencies. Safe.

Self-circularity: The bound $\sigma_A \leq 1$ is not strictly proved by the ODE alone for all parameter regimes. The proof must be supplemented by the equilibrium analysis or the numerical clamp. This is a **partial circular dependency**: Proposition 3.2’s claim of $\sigma_A \leq 1$ relies on Proposition 3.4’s equilibrium bound or Algorithm 3.1’s clamping. We flag this for the reader.

4 Fast-Slow Decomposition (Fenichel)

Definition 4.1 (Fast and Slow Variables). *Decompose the state vector:*

$$\mathbf{x}_A^{fast} = (\delta_A, \sigma_A, \alpha_A) \in \mathbb{R}^3, \quad \mathbf{x}_A^{slow} = (\hat{M}_A, \Xi_A^P, \Xi_A^I, \Xi_A^F) \in \mathbb{R}^4.$$

Definition 4.2 (Timescale Parameter). *Define the timescale separation parameter:*

$$\epsilon \triangleq \frac{\max(\nu_M, \kappa_P, \kappa_I, \kappa_F)}{\min(\eta_{\max}, \rho, \gamma)}.$$

The original manuscript asserts $\epsilon \ll 1$ as a design condition. We treat it as an assumption for the validity of the decomposition.

Assumption 4.3 (Timescale Separation). *The parameters satisfy:*

$$\epsilon \ll 1.$$

That is, the slow subsystem’s rate constants ($\nu_M \sim 10^{-2}$, $\kappa. \sim 10^{-2}$) are at least one order of magnitude smaller than the fast subsystem’s rates ($\eta_{\max} \sim 1$, $\rho \sim 10^{-1}$, $\gamma \sim 10^{-1}$).

Assumption 4.4 (Critical Manifold Compactness). *The critical manifold $\mathcal{M}_0 = \{\mathbf{x}_A \in \mathcal{X} : \dot{\mathbf{x}}_A^{fast} = 0\}$ is compact and connected. This follows from Proposition 3.2: $\mathcal{X}$ is forward-invariant and compact, and $\mathcal{M}_0$ is a closed subset of $\mathcal{X}$.*

Proposition 4.5 (Fast-Slow Decomposition). *Under Assumptions 4.3–2, the system admits a slow manifold $\mathcal{M}_\epsilon = \{\mathbf{x}_A : \mathbf{x}_A^{fast} = \mathbf{h}(\mathbf{x}_A^{slow}, \epsilon)\}$ that is $O(\epsilon)$ -close to the critical manifold $\mathcal{M}_0 = \{\mathbf{x}_A : \dot{\mathbf{x}}_A^{fast} = 0\}$, and the reduced dynamics on $\mathcal{M}_\epsilon$ follow:*

$$\dot{\mathbf{x}}_A^{slow} = \epsilon \cdot \mathbf{g}(\mathbf{h}(\mathbf{x}_A^{slow}, \epsilon), \mathbf{x}_A^{slow}).$$

Proof. Fenichel’s geometric singular perturbation theory (Fenichel, 1979) requires three conditions: (i) the critical manifold $\mathcal{M}_0$ is compact and connected, (ii) the fast subsystem’s linearization restricted to the normal bundle of $\mathcal{M}_0$ has no eigenvalues with zero real part (normal hyperbolicity), and (iii) the vector field is C^r for sufficiently large r . We verify each below.

Condition (i): Compactness and connectedness. By Assumption 2, $\mathcal{M}_0$ is a closed subset of the compact set $\mathcal{X}$ under Proposition 3.2; hence $\mathcal{M}_0$ is compact. Connectedness follows from the continuity of the fixed-point equation $\dot{\mathbf{x}}_A^{fast} = 0$: on each connected component of $\mathcal{X}$ (which is a product of intervals), the solution set of analytic equations is connected.

Condition (ii): Normal hyperbolicity. The original manuscript (line 923) claims the Jacobian of the fast subsystem “has eigenvalues with strictly negative real parts (due to the saturating nonlinearities in the Gompertz efficiency and the $(1 - \sigma_A)$, $(1 - \alpha_A)$ factors).” **This is a hand-wave that must be replaced by an explicit calculation.**

The fast subsystem Jacobian is $D\mathbf{f}_{fast} \in \mathbb{R}^{3 \times 3}$:

$$D\mathbf{f}_{fast} = \begin{pmatrix} \frac{\partial \dot{\delta}_A}{\partial \delta_A} & \frac{\partial \dot{\delta}_A}{\partial \sigma_A} & \frac{\partial \dot{\delta}_A}{\partial \alpha_A} \\ \frac{\partial \dot{\sigma}_A}{\partial \delta_A} & \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} & \frac{\partial \dot{\sigma}_A}{\partial \alpha_A} \\ \frac{\partial \dot{\alpha}_A}{\partial \delta_A} & \frac{\partial \dot{\alpha}_A}{\partial \sigma_A} & \frac{\partial \dot{\alpha}_A}{\partial \alpha_A} \end{pmatrix}.$$

We compute each entry explicitly at a point on $\mathcal{M}_0$.

α_A **submatrix (fastest variable).**

Recall:

$$\dot{\alpha}_A = \gamma C_A(1 - \alpha_A) - \zeta_\alpha \alpha_A R_A^{\text{surface}}.$$

The self-derivative is:

$$\frac{\partial \dot{\alpha}_A}{\partial \alpha_A} = -\gamma C_A - \zeta_\alpha R_A^{\text{surface}}.$$

Since $\gamma, C_A, \zeta_\alpha, R_A^{\text{surface}} \geq 0$, and typically $\gamma C_A > 0$ (Assumption $C_A > 0$), we have:

$$\frac{\partial \dot{\alpha}_A}{\partial \alpha_A} = -\underbrace{(\gamma C_A + \zeta_\alpha R_A^{\text{surface}})}_{>0} < 0.$$

The cross-derivatives:

$$\frac{\partial \dot{\alpha}_A}{\partial \delta_A} = \gamma(1 - \alpha_A) \frac{\partial C_A}{\partial \delta_A} - \zeta_\alpha \alpha_A \frac{\partial R_A^{\text{surface}}}{\partial \delta_A}, \quad \frac{\partial \dot{\alpha}_A}{\partial \sigma_A} = 0$$

(C_A and R_A^{surface} do not depend directly on σ_A).

σ_A **submatrix.**

Recall:

$$\dot{\sigma}_A = \rho P_A \alpha_A \sigma_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL} \sigma_A.$$

The self-derivative is:

$$\frac{\partial \dot{\sigma}_A}{\partial \sigma_A} = \rho P_A \alpha_A (1 - 2\gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}.$$

On the critical manifold $\mathcal{M}_0$ (where $\dot{\sigma}_A = 0$):

$$\rho P_A \alpha_A \sigma_A^* (1 - \gamma_\sigma \sigma_A^*) = \epsilon_\sigma \Omega_{SL} \sigma_A^*.$$

For $\sigma_A^* > 0$, dividing by σ_A^* :

$$\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A^*) = \epsilon_\sigma \Omega_{SL}.$$

Substituting into the self-derivative:

$$\left. \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} \right|_{\mathcal{M}_0} = \rho P_A \alpha_A (1 - 2\gamma_\sigma \sigma_A^*) - \underbrace{\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A^*)}_{\epsilon_\sigma \Omega_{SL}} = -\rho P_A \alpha_A \gamma_\sigma \sigma_A^* < 0.$$

For $\sigma_A^* = 0$:

$$\left. \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} \right|_{\sigma_A=0} = \rho P_A \alpha_A - \epsilon_\sigma \Omega_{SL} = \epsilon_\sigma \Omega_{SL} (R_0 - 1).$$

This is negative when $R_0 < 1$ (schema-free regime) and positive when $R_0 > 1$ (the bifurcation condition). The $\sigma_A = 0$ branch of $\mathcal{M}_0$ loses normal hyperbolicity at $R_0 = 1$ — the transcritical bifurcation point studied in Proposition 4.1. For $R_0 \neq 1$, $\partial \dot{\sigma}_A / \partial \sigma_A \neq 0$ on the $\sigma_A = 0$ branch.

The remaining cross-derivatives:

$$\frac{\partial \dot{\sigma}_A}{\partial \alpha_A} = \rho P_A \sigma_A (1 - \gamma_\sigma \sigma_A) \geq 0, \quad \frac{\partial \dot{\sigma}_A}{\partial \delta_A} = \rho \sigma_A (1 - \gamma_\sigma \sigma_A) \left(\alpha_A \frac{\partial P_A}{\partial \delta_A} + P_A \frac{\partial \alpha_A}{\partial \delta_A} \right) - \epsilon_\sigma \sigma_A \frac{\partial \Omega_{SL}}{\partial \delta_A}.$$

δ_A submatrix.

Recall:

$$\dot{\delta}_A = f_{\text{learn}} \eta(\delta_A) T_A - \lambda_c (1 - \gamma_\sigma \sigma_A) \delta_A (1 - r_A),$$

where $\eta(\delta_A) = \eta_{\text{max}} \exp(-a \exp(-b \delta_A^{\text{relative}}))$ is the Gompertz efficiency. The self-derivative:

$$\frac{\partial \dot{\delta}_A}{\partial \delta_A} = f_{\text{learn}} T_A \frac{\partial \eta}{\partial \delta_A} - \lambda_c (1 - \gamma_\sigma \sigma_A) (1 - r_A).$$

The Gompertz derivative:

$$\frac{\partial \eta}{\partial \delta_A} = \eta \cdot ab \exp(-b \delta_A^{\text{relative}}) \cdot \frac{1}{\Delta(d, t)} > 0$$

for finite $\delta_A < \Delta$. The decay term coefficient $-\lambda_c (1 - \gamma_\sigma \sigma_A) (1 - r_A)$ is negative. The sign of $\partial \dot{\delta}_A / \partial \delta_A$ depends on the balance; near equilibrium $\dot{\delta}_A = 0$, the decay term dominates small perturbations, yielding negativity.

Cross-derivatives:

$$\frac{\partial \dot{\delta}_A}{\partial \sigma_A} = \lambda_c \gamma_\sigma \delta_A (1 - r_A) \geq 0, \quad \frac{\partial \dot{\delta}_A}{\partial \alpha_A} = 0$$

(no direct α_A dependence in the δ_A ODE).

Condition (ii) conclusion. The Jacobian on $\mathcal{M}_0$ (away from $R_0 = 1$) has the block-triangular structure:

$$D\mathbf{f}_{\text{fast}} = \begin{pmatrix} \partial_{\delta_A} \dot{\delta}_A & \partial_{\sigma_A} \dot{\delta}_A & 0 \\ \partial_{\delta_A} \dot{\sigma}_A & \partial_{\sigma_A} \dot{\sigma}_A & \partial_{\alpha_A} \dot{\sigma}_A \\ \partial_{\delta_A} \dot{\alpha}_A & 0 & \partial_{\alpha_A} \dot{\alpha}_A \end{pmatrix}.$$

The eigenvalues are the diagonal entries of the Schur complement after block diagonalization:

$$\lambda_1 = \frac{\partial \dot{\alpha}_A}{\partial \alpha_A} = -(\gamma C_A + \zeta_\alpha R_A^{\text{surface}}) < 0,$$

λ_2, λ_3 = eigenvalues of the 2×2 upper-left block.

The 2×2 block:

$$J_{2 \times 2} = \begin{pmatrix} \partial_{\delta_A} \dot{\delta}_A & \partial_{\sigma_A} \dot{\delta}_A \\ \partial_{\delta_A} \dot{\sigma}_A & \partial_{\sigma_A} \dot{\sigma}_A \end{pmatrix}.$$

Its trace:

$$\text{Tr}(J_{2 \times 2}) = \frac{\partial \dot{\delta}_A}{\partial \delta_A} + \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} < 0 \quad (\text{both diagonal entries negative on } \mathcal{M}_0 \text{ away from } R_0 = 1).$$

Its determinant:

$$\det(J_{2 \times 2}) = \frac{\partial \dot{\delta}_A}{\partial \delta_A} \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} - \frac{\partial \dot{\delta}_A}{\partial \sigma_A} \frac{\partial \dot{\sigma}_A}{\partial \delta_A}.$$

Both $\partial \dot{\delta}_A / \partial \delta_A$ and $\partial \dot{\sigma}_A / \partial \sigma_A$ are negative, and the product of the cross-derivatives $\partial \dot{\delta}_A / \partial \sigma_A \cdot \partial \dot{\sigma}_A / \partial \delta_A$ is typically second-order small (each involves σ_A or a derivative of coefficient functions). Hence $\det(J_{2 \times 2}) > 0$ on $\mathcal{M}_0$. Since trace < 0 and determinant > 0 , both eigenvalues have strictly negative real parts.

Therefore, all three eigenvalues of $D\mathbf{f}_{\text{fast}}$ on $\mathcal{M}_0$ (away from $R_0 = 1$) have strictly negative real parts. Normal hyperbolicity holds. At $R_0 = 1$, a simple eigenvalue crosses zero — this is the transcritical bifurcation barrier where Fenichel's theorem does not apply and the decomposition breaks down (the exchange lemma handles this case).

Condition (iii): Smoothness. By Proposition 3.1, the vector field is piecewise- C^∞ with finitely many smooth components separated by measure-zero discontinuity sets. Fenichel’s theorem requires C^r for $r \geq 1$; the piecewise- C^∞ structure satisfies this on each component. The non-differentiable points ($\max(0, \cdot)$, Corr singularities) are confined to lower-dimensional submanifolds that do not intersect $\mathcal{M}_0$ in general position (the critical manifold lies in the interior of the smooth regime for typical parameters).

Application of Fenichel’s theorem. With conditions (i)–(iii) satisfied, Fenichel’s theorem guarantees:

1. Persistence of $\mathcal{M}_0$ to $\mathcal{M}_\epsilon$ for sufficiently small ϵ .
2. $\mathcal{M}_\epsilon$ is $O(\epsilon)$ -close to $\mathcal{M}_0$.
3. The reduced dynamics on $\mathcal{M}_\epsilon$ are given by $\dot{\mathbf{x}}_A^{\text{slow}} = \epsilon \cdot \mathbf{g}(\mathbf{h}(\mathbf{x}_A^{\text{slow}}, \epsilon), \mathbf{x}_A^{\text{slow}})$.

□

□

Boundary Case Verification. $R_0 = 1$ **exactly:** At the bifurcation point, $\partial \dot{\sigma}_A / \partial \sigma_A = 0$, violating normal hyperbolicity. Fenichel’s theorem does not apply. The fast-slow decomposition must be replaced by the exchange lemma or center manifold reduction near this value. This is physically interpretable as the Phase 1 $\rightarrow$ 2 transition region, where the system spends arbitrarily long time near the bifurcation point (critical slowing down).

$\epsilon \approx 1$ (**no separation**): If the rate constants are comparable, the decomposition is invalid. The paper relies on $\epsilon \ll 1$ as an assumption derived from order-of-magnitude estimates, but this has not been empirically verified.

$\sigma_A^* = 0$ **branch:** For $R_0 < 1$, the $\sigma_A = 0$ branch of $\mathcal{M}_0$ is normally hyperbolic. For $R_0 > 1$, it loses stability and the $\sigma_A^* > 0$ branch becomes the attracting slow manifold.

Circular Dependency Check. **Does Proposition 3.3 depend on any later result?** It depends on Proposition 3.2 (for compactness of $\mathcal{X}$ and forward invariance) and Proposition 3.1 (for existence of solutions). These are forward dependencies.

It also depends on Proposition 4.1 for the characterization at $R_0 = 1$, but this is an edge case — the primary claim holds for $R_0 \neq 1$ independently.

Does any later result depend on Proposition 3.3? The equilibrium analysis (Proposition 3.4) and bifurcation analysis (Proposition 4.1) benefit from the decomposition but do not strictly depend on it. The numerical integration scheme (Algorithm 3.1’s IMEX split) is motivated by Proposition 3.3 but can be used even without the rigorous decomposition. Safe.

5 Single-Domain Equilibria

Definition 5.1 (Single-Domain State Space). When $|D| = 1$, denote $\sigma_A = \sigma_A(d, \cdot)$, $\delta_A = \delta_A(d, \cdot)$, $\alpha_A = \alpha_A(d, \cdot)$.

Assumption 5.2 (Symmetric Couplings). All cross-domain coupling terms vanish when $|D| = 1$: no intersection discovery, no interdomain schema transfer, no multi-domain attention.

Proposition 5.3 (Single-Domain Equilibria). Under the single-domain reduction (Assumption 1), the system has precisely two equilibrium families:

1. **Schema-free equilibrium:** $\sigma_A = 0$, $\delta_A = 0$, $\alpha_A = \gamma C_A / (\gamma C_A + \zeta_\alpha R_A^{\text{surface}})$. Exists for all parameter values.
2. **Schema-active equilibrium:** $\sigma_A^* > 0$ defined in Proposition 4.1, δ_A^* the positive solution of $\dot{\delta}_A = 0$, α_A^* the positive solution of $\dot{\alpha}_A = 0$. Exists iff $R_0 \geq 1$.

Proof. Setting $\dot{\sigma}_A = 0$ gives $\sigma_A = 0$ or $\sigma_A^* = (1 - 1/R_0)/\gamma_\sigma$ (Proposition 4.1). Setting $\dot{\alpha}_A = 0$ yields:

$$\alpha_A^* = \frac{\gamma C_A}{\gamma C_A + \zeta_\alpha R_A^{\text{surface}}} \in (0, 1).$$

Setting $\dot{\delta}_A = 0$ at either equilibrium defines δ_A^* implicitly via:

$$f_{\text{learn}}\eta(\delta_A)T_A = \lambda_c(1 - \gamma_\sigma\sigma_A)\delta_A(1 - r_A).$$

At $\sigma_A = 0$, the decay coefficient is λ_c ; at σ_A^* , the decay coefficient is $\lambda_c(1 - \gamma_\sigma\sigma_A^*) < \lambda_c$, meaning schema-active agents retain depth more effectively. $\square$

Part III

Estimation and Numerical Properties

6 Estimator Consistency

Definition 6.1 (Two-Stage Estimation). *Let $\tilde{\sigma}_A(d, t)$ be the Stage 1 (training-time) proxy computed by fusing three structural signals:*

- **GCA** (Generalised Compositional Alignment): *correlation between structural similarity and performance difference,*
- **RGA** (Relational Generalisation Alignment): *rank correlation of compositional generalisation patterns,*
- **AC** (Augmentation Consistency): *agreement under compositional input transformations.*

Let $\hat{\sigma}_A(d, t) = \text{Acc}_{\text{OOD}}/\text{Acc}_{\text{ID}}$ be the Stage 2 (evaluation-time) ground-truth estimate. The discrepancy is

$$\delta_\sigma(d, t) = \tilde{\sigma}_A(d, t) - \hat{\sigma}_A(d, t).$$

Definition 6.2 (Generative Grammar $G(d)$). *A probabilistic context-free grammar that generates compositional structures for domain d . The recombination distribution $\mathcal{R}(G(d))$ governs how primitive elements combine into probe samples used in Stage 1 estimation.*

Assumption 6.3 (Correct Generative Model). *The generative grammar $G(d)$ correctly models the domain’s compositional structure. The compositional probe batch $\{s_j\}_{j=1}^{n_{\text{batch}}}$ is drawn i.i.d. from the recombination distribution $\mathcal{R}(G(d))$.*

Assumption 6.4 (Bounded Signals). *Each structural signal is bounded in $[0, 1]$ and has finite variance. The fusion weights are non-negative and sum to 1.*

Proposition 6.5 (Estimator Consistency). *Under Assumptions 1–2, the multi-signal fusion estimator satisfies:*

$$\mathbb{E}[|\delta_\sigma|] \leq \frac{C_1}{\sqrt{n_{\text{batch}}}} + C_2 \cdot \epsilon_{\text{model}},$$

where n_{batch} is the number of compositional probe samples, ϵ_{model} is the model misspecification error, and C_1, C_2 are constants depending on the domain parameters.

Proof. We decompose the expected discrepancy into statistical and systematic components:

$$\mathbb{E}[|\delta_\sigma|] \leq \underbrace{\mathbb{E}[|\tilde{\sigma}_A - \mathbb{E}[\tilde{\sigma}_A]|]}_{\text{statistical}} + \underbrace{|\mathbb{E}[\tilde{\sigma}_A] - \sigma_A^*|}_{\text{systematic}} + \underbrace{|\sigma_A^* - \hat{\sigma}_A|}_{\text{measurement}},$$

where σ_A^* is the true schema coherence. The third term is absorbed into ϵ_{model} as it represents fundamental unidentifiability.

Step 1: Statistical term — concentration of the fusion estimate. The Stage 1 proxy is a weighted combination:

$$\tilde{\sigma}_A = w_1 \cdot \text{GCA} + w_2 \cdot \text{RGA} + w_3 \cdot \text{AC}.$$

Each signal is a U-statistic or sample mean computed from the compositional probe batch.

For the correlation-based signals (GCA, RGA), each is a degree- m U-statistic:

$$U = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} h(X_{i_1}, \dots, X_{i_m}),$$

where h is a symmetric kernel bounded in $[0, 1]$. By Hoeffding’s inequality for U-statistics (Hoeffding, 1948):

$$\mathbb{P}(|U - \mathbb{E}[U]| \geq t) \leq 2 \exp(-2 \lfloor n/m \rfloor t^2).$$

For the augmentation consistency AC, which is a sample mean of bounded indicators, Hoeffding’s inequality gives:

$$\mathbb{P}(|\text{AC} - \mathbb{E}[\text{AC}]| \geq t) \leq 2 \exp(-2 n_{\text{batch}} t^2).$$

Applying a union bound over the three signals with $t = \varepsilon/3$:

$$\mathbb{P}(|\tilde{\sigma}_A - \mathbb{E}[\tilde{\sigma}_A]| \geq \varepsilon) \leq 6 \exp(-2 \lfloor n_{\text{batch}}/m \rfloor (\varepsilon/3)^2).$$

Integrating the tail bound gives:

$$\mathbb{E}[|\tilde{\sigma}_A - \mathbb{E}[\tilde{\sigma}_A]|] = \int_0^\infty \mathbb{P}(|\tilde{\sigma}_A - \mathbb{E}[\tilde{\sigma}_A]| \geq t) dt \leq \frac{C_1}{\sqrt{n_{\text{batch}}}},$$

where $C_1 = 3\sqrt{2m}$ (absorbing constants from the Gaussian tail integral of the Hoeffding bound).

Step 2: Systematic term — model misspecification. The systematic bias $|\mathbb{E}[\tilde{\sigma}_A] - \sigma_A^*|$ arises from two sources:

1. **Fusion weight misspecification:** The fixed weights (w_1, w_2, w_3) may not be optimal for the current domain.
2. **Grammar approximation error:** $G(d)$ may not capture all compositional structure, inducing a bias in the probe distribution.

Both sources are bounded by a domain-specific constant C_2 times the model misspecification error ϵ_{model} , defined as the total variation distance between the true recombination distribution and the grammar-induced distribution.

Step 3: Measurement term — OOD/ID ratio consistency. The Stage 2 estimator $\hat{\sigma}_A = \text{Acc}_{\text{OOD}}/\text{Acc}_{\text{ID}}$ is a ratio of two bounded random variables. When $\text{Acc}_{\text{ID}} > 0$ (non-degenerate case), the Delta method gives:

$$|\sigma_A^* - \hat{\sigma}_A| \leq \frac{|\text{Acc}_{\text{OOD}} - \mathbb{E}[\text{Acc}_{\text{OOD}}]|}{\mathbb{E}[\text{Acc}_{\text{ID}}]} + \frac{\mathbb{E}[\text{Acc}_{\text{OOD}}] \cdot |\text{Acc}_{\text{ID}} - \mathbb{E}[\text{Acc}_{\text{ID}}]|}{(\mathbb{E}[\text{Acc}_{\text{ID}}])^2} + o_p(1).$$

Each accuracy estimate concentrates at rate $1/\sqrt{n_{\text{eval}}}$. For $n_{\text{eval}} \geq n_{\text{batch}}$, this term is dominated by the statistical term from Step 1.

Step 4: Combining terms. Taking expectations and collecting constants:

$$\mathbb{E}[|\delta_\sigma|] \leq \frac{3\sqrt{2m}}{\sqrt{n_{\text{batch}}}} + C_2 \cdot \epsilon_{\text{model}}.$$

Setting $C_1 = 3\sqrt{2m}$ yields the stated bound. $\square$

Boundary Case Verification. $n_{\text{batch}} \rightarrow \infty$: The statistical term vanishes and $\mathbb{E}[|\delta_\sigma|] \rightarrow C_2 \epsilon_{\text{model}}$. Even with infinite data, systematic misspecification persists.

$\epsilon_{\text{model}} = 0$: If $G(d)$ is exact, $\mathbb{E}[|\delta_\sigma|] \leq C_1/\sqrt{n_{\text{batch}}}$, decaying to zero as $n_{\text{batch}} \rightarrow \infty$.

$\text{AccID} = 0$: The ratio $\hat{\sigma}_A$ is undefined (division by zero). This occurs when the agent has zero accuracy on ID tasks, indicating catastrophic forgetting. The estimator is undefined and the recalibration rule triggers a diagnostic.

Discrepancy > 0.15 : The recalibration rule (ridge regression refit of fusion weights) is triggered. The systematic bias term $C_2 \epsilon_{\text{model}}$ is reduced, at the cost of introducing finite-sample noise from the smaller recalibration dataset.

Circular Dependency Check. **Dependencies on earlier results:** The proof uses standard concentration inequalities and does not depend on any earlier proposition. It assumes the ODE state variables are well-defined (Section 3.1) but does not require Proposition 3.1 or 3.2.

Forward dependencies: Proposition 3.7 (error propagation) depends on this proposition to bound $|\delta_\sigma|$. This is a forward dependency, not circular.

Self-circularity: The bound $C_1 = 3\sqrt{2m}$ requires m , the degree of the U-statistic kernel. For GCA (Pearson correlation), $m = 2$; for RGA (Spearman rank correlation), $m = 2$; for AC (sample mean), $m = 1$. A conservative bound takes $m = 2$, giving $C_1 = 6$. The constant C_2 depends on the domain and is not explicitly computed; its existence follows from the boundedness of the fusion estimator and the total variation bound on the grammar approximation error.

7 Error Propagation

Definition 7.1 (Estimation Error). Let $\delta_\sigma(d, t) = \tilde{\sigma}_A(d, t) - \hat{\sigma}_A(d, t)$ be the Stage 1–Stage 2 discrepancy (Proposition 3.6). The true schema coherence is $\sigma_A^*(d, t)$; the operative value used in the ODE is $\tilde{\sigma}_A(d, t)$.

Definition 7.2 (Propagated ODE Error). Let $\dot{\delta}_A^{\text{error}}(d, t)$ be the deviation in the depth and schema ODE right-hand sides caused by using $\tilde{\sigma}_A$ instead of σ_A^* :

$$\dot{\delta}_A^{\text{error}} = |f(\mathbf{x}; \tilde{\sigma}_A) - f(\mathbf{x}; \sigma_A^*)|,$$

where f denotes the vector field of the coupled ODE system.

Assumption 7.3 (Lipschitz Continuity of Coupling Terms). The coupling functions $\sigma_A \mapsto \lambda_c \gamma_\sigma \sigma_A \delta_A$ and $\sigma_A \mapsto \rho P_A \alpha_A \sigma_A$ are Lipschitz continuous with constants $L_1 = \lambda_c \gamma_\sigma \delta_A$ and $L_2 = \rho P_A \alpha_A$, respectively.

Assumption 7.4 (Bounded Variables). All state variables are bounded as established in Proposition 3.2: $\sigma_A \in [0, 1]$, $\delta_A \in [0, \Delta_{\text{max}}]$, $\alpha_A \in [0, 1]$, $P_A \in [0, P_{\text{max}}]$.

Proposition 7.5 (Error Propagation in ODE System). Under Assumptions 1–2, the effective error propagated into the depth ODE (??) and schema ODE (??) is bounded by:

$$|\dot{\delta}_A^{\text{error}}| \leq \lambda_c \cdot \gamma_\sigma \cdot |\delta_\sigma| \cdot \delta_A + \rho \cdot P_A \cdot \alpha_A \cdot |\delta_\sigma|.$$

If $|\delta_\sigma| < 0.10$, the propagated error is negligible relative to typical ODE step sizes ($\eta_{\text{max}} \sim O(1)$).

Proof. We trace the two pathways through which σ_A enters the coupled ODE system.

Step 1: Depth ODE perturbation. The depth ODE (??) is:

$$\dot{\delta}_A = f_{\text{learn}} \cdot \eta \cdot T_A - \lambda_c \cdot (1 - \gamma_\sigma \cdot \sigma_A) \cdot \delta_A \cdot (1 - r_A).$$

The only σ_A -dependent term is the decay component $-\lambda_c(1 - \gamma_\sigma \sigma_A)\delta_A(1 - r_A)$. Let $\tilde{\sigma}_A = \sigma_A^* + \delta_\sigma$ be the corrupted value. The deviation in the depth ODE is:

$$\begin{aligned} \Delta_\delta &= |[-\lambda_c(1 - \gamma_\sigma \tilde{\sigma}_A)\delta_A(1 - r_A)] - [-\lambda_c(1 - \gamma_\sigma \sigma_A^*)\delta_A(1 - r_A)]| \\ &= \lambda_c \delta_A(1 - r_A) \cdot |\gamma_\sigma \tilde{\sigma}_A - \gamma_\sigma \sigma_A^*| \\ &= \lambda_c \gamma_\sigma \delta_A(1 - r_A) \cdot |\delta_\sigma|. \end{aligned}$$

Since $(1 - r_A) \in [0, 1]$, we have:

$$\Delta_\delta \leq \lambda_c \gamma_\sigma \delta_A |\delta_\sigma|.$$

This is the first term in the bound.

Step 2: Schema ODE perturbation. The schema ODE (??) is:

$$\dot{\sigma}_A = \sigma_A \cdot [\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}].$$

The σ_A factor outside the bracket is the operative value (the state variable itself, not the estimate). The error propagation occurs through the σ_A inside the bracket — specifically, the $(1 - \gamma_\sigma \sigma_A)$ which should use σ_A^* .

Let $\tilde{\sigma}_A = \sigma_A^* + \delta_\sigma$ be used in the bracket's $(1 - \gamma_\sigma \sigma_A)$ term. The deviation in the schema ODE at a given state is:

$$\begin{aligned} \Delta_\sigma &= |\sigma_A [\rho P_A \alpha_A (1 - \gamma_\sigma \tilde{\sigma}_A) - \epsilon_\sigma \Omega_{SL}] - \sigma_A [\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A^*) - \epsilon_\sigma \Omega_{SL}]| \\ &= \sigma_A \cdot \rho P_A \alpha_A \cdot |\gamma_\sigma \tilde{\sigma}_A - \gamma_\sigma \sigma_A^*| \\ &= \sigma_A \cdot \rho P_A \alpha_A \gamma_\sigma \cdot |\delta_\sigma|. \end{aligned}$$

Using $\sigma_A \leq 1$ and $\gamma_\sigma \leq 1$:

$$\Delta_\sigma \leq \rho P_A \alpha_A |\delta_\sigma|.$$

This is the second term in the bound.

Step 3: Combining terms. The total propagated error magnitude is the sum of the two pathways:

$$|\dot{\delta}_A^{\text{error}}| = \Delta_\delta + \Delta_\sigma \leq \lambda_c \gamma_\sigma \delta_A |\delta_\sigma| + \rho P_A \alpha_A |\delta_\sigma|,$$

which is the stated inequality.

Step 4: Negligibility condition. The typical ODE step size is determined by the fastest rate constant $\eta_{\max} \sim O(1)$. The propagated error is negligible when:

$$\lambda_c \gamma_\sigma \delta_A |\delta_\sigma| + \rho P_A \alpha_A |\delta_\sigma| \ll \eta_{\max}.$$

Using the numerical values $\lambda_c = 0.01$, $\gamma_\sigma \in (0, 1)$, $\rho = 0.1$, $\alpha_A \leq 1$, $P_A \leq 1$, $\delta_A \leq 1$:

$$\lambda_c \gamma_\sigma \delta_A |\delta_\sigma| \leq 0.01 |\delta_\sigma|, \quad \rho P_A \alpha_A |\delta_\sigma| \leq 0.1 |\delta_\sigma|.$$

The total is $\leq 0.11 |\delta_\sigma|$. For $|\delta_\sigma| < 0.10$:

$$|\dot{\delta}_A^{\text{error}}| \leq 0.011 \ll 1 \approx \eta_{\max}.$$

Thus the propagated error is at most 1.1% of the fastest rate, rendering it negligible for numerical integration and qualitative dynamics. $\square$

Boundary Case Verification. $\delta_A = 0$ (**nascent domain**): The depth ODE perturbation term vanishes ($\Delta_\delta = 0$). Only the schema ODE perturbation remains, bounded by $\rho P_A \alpha_A |\delta_\sigma|$.

$\alpha_A = 0$ (**attention collapse**): Both perturbation terms vanish. Errors in σ_A estimation do not propagate when there is no attentional gating. This is consistent: if $\alpha_A = 0$, the ODE dynamics are already frozen.

$P_A = 0$ (**no prior knowledge**): The schema perturbation term vanishes. Only depth perturbation, bounded by $\lambda_c \gamma_\sigma \delta_A |\delta_\sigma|$, remains.

$|\delta_\sigma| \geq 0.10$: The propagated error may become non-negligible. The recalibration rule (ridge regression, Proposition 3.6) is triggered to reduce $|\delta_\sigma|$ below this threshold, restoring negligibility.

Circular Dependency Check. **Dependencies on earlier results:** The proof uses Proposition 3.6 to define δ_σ and its bound. It also uses the ODE definitions from Section 3.2 and Proposition 3.2 (boundedness of state variables). These are all earlier results.

Forward dependencies: The error bound is used implicitly in Theorem 4.1 to ensure that SGD suppression analysis is robust to estimation noise. This is a forward dependency, not circular.

Self-circularity: The bound assumes the error propagates additively through the two pathways. In reality, the ODE is coupled and errors can amplify through feedback — δ_σ affects $\dot{\sigma}_A$, which changes $\sigma_A(t + \Delta t)$, which changes δ_σ at the next timestep. The stated bound is a one-step bound (instantaneous), not a global bound over a time interval. A full global error bound would require a Gronwall-type argument integrating the Lipschitz constant over time. The one-step bound is sufficient for the claim’s purpose (assessing negligibility relative to $\eta_{\max}$).

8 Numerical Stability

Definition 8.1 (ODE System Jacobian). Let $\mathbf{x}_A(t) \in \mathbb{R}^N$ be the full state vector (??) with $N = |D| \times |M| \times 4 + 3$. The system Jacobian $\mathbf{J} \in \mathbb{R}^{N \times N}$ is:

$$\mathbf{J}_{ij} = \frac{\partial \dot{\mathbf{x}}_{A,i}}{\partial \mathbf{x}_{A,j}}.$$

The condition number is $\kappa(\mathbf{J}) = \|\mathbf{J}\| \cdot \|\mathbf{J}^{-1}\|$ for any induced matrix norm.

Definition 8.2 (IMEX Partitioning). Algorithm 3.1 partitions the state vector into:

- **Fast subsystem** $\mathbf{x}_F = (\delta_A, \sigma_A, \alpha_A)$: rate constants $\eta_{\max} \sim O(1)$, $\rho \sim O(10^{-1})$, $\gamma \sim O(10^{-1})$.
- **Slow subsystem** $\mathbf{x}_S = (\hat{M}_A, \Xi_A^P, \Xi_A^I, \Xi_A^F)$: rate constants $\nu_M, \kappa_P, \kappa_I, \kappa_F \sim O(10^{-2})$.

Assumption 8.3 (Boundedness of State Variables). All state variables lie in the compact forward-invariant set $\mathcal{X}$ (Proposition 3.2). In particular, $\sigma_A, \alpha_A \in [0, 1]$ and all other variables are $O(1)$ bounded.

Proposition 8.4 (Numerical Stability). Under Assumption 1, the condition number of the Jacobian satisfies:

$$\kappa(\mathbf{J}) \leq C \cdot \frac{\max(\eta_{\max}, \rho, \gamma)}{\min(\nu_M, \kappa_P, \kappa_I, \kappa_F)} \cdot \frac{1}{1 - \max(\sigma_A, \alpha_A)},$$

where C is a constant depending on the domain parameters $|D|$ and $|M|$. The condition number diverges as $\sigma_A \rightarrow 1$ or $\alpha_A \rightarrow 1$, indicating numerical stiffness near saturation.

Proof. The Jacobian $\mathbf{J}$ has a block structure induced by the ODE coupling:

$$\mathbf{J} = \begin{pmatrix} \mathbf{J}_{FF} & \mathbf{J}_{FS} \\ \mathbf{J}_{SF} & \mathbf{J}_{SS} \end{pmatrix},$$

where $\mathbf{J}_{FF}$ couples the fast variables, $\mathbf{J}_{SS}$ couples the slow variables, and the off-diagonal blocks represent cross-coupling.

Step 1: Spectral bound on the fast block. The fast subsystem ODEs are:

$$\begin{aligned}\dot{\delta}_A &= f_{\text{learn}} \cdot \eta_{\text{max}} e^{-ae^{-b\delta_A}} \cdot T_A - \lambda_c(1 - \gamma_\sigma \sigma_A) \delta_A(1 - r_A), \\ \dot{\sigma}_A &= \sigma_A [\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}], \\ \dot{\alpha}_A &= \gamma C_A (1 - \alpha_A) - \zeta_\alpha \alpha_A R_A^{\text{surface}}.\end{aligned}$$

The partial derivatives $\partial \dot{\sigma}_A / \partial \sigma_A$ and $\partial \dot{\alpha}_A / \partial \alpha_A$ contain factors $(1 - 2\gamma_\sigma \sigma_A)$ and $(1 - 2\alpha_A)$ respectively, which approach zero as $\sigma_A, \alpha_A \rightarrow 1$. In the saturation limit, the self-coupling vanishes, making those rows/columns nearly zero. By Gershgorin's circle theorem, the eigenvalues of $\mathbf{J}_{FF}$ are bounded by:

$$|\lambda(\mathbf{J}_{FF})| \leq \max(\eta_{\text{max}}, \rho, \gamma) + O(1).$$

The $O(1)$ additive term comes from bounded coupling terms ($\lambda_c, f_{\text{learn}}, \zeta_\alpha$, etc.).

Step 2: Spectral bound on the slow block. The slow subsystem variables ($\hat{M}_A, \Xi_A$) have dynamics with negative self-feedback at rate constants $\nu_M, \kappa_P, \kappa_I, \kappa_F \sim O(10^{-2})$. The eigenvalues of $\mathbf{J}_{SS}$ satisfy:

$$|\lambda(\mathbf{J}_{SS})| \geq c_S \cdot \min(\nu_M, \kappa_P, \kappa_I, \kappa_F)$$

for some constant $c_S > 0$, because the self-coupling terms dominate the bounded off-diagonal couplings.

Step 3: Block matrix condition number bound. For a block triangularizable matrix, the condition number satisfies (Weyl's inequality for block LU factorization):

$$\kappa(\mathbf{J}) \leq \frac{\max_i |\lambda_i(\mathbf{J})|}{\min_i |\lambda_i(\mathbf{J})|}.$$

The maximum eigenvalue magnitude is bounded by the spectral radius of $\mathbf{J}_{FF}$:

$$\max_i |\lambda_i(\mathbf{J})| \leq C_1 \cdot \max(\eta_{\text{max}}, \rho, \gamma) \cdot \frac{1}{1 - \max(\sigma_A, \alpha_A)},$$

where $(1 - \max(\sigma_A, \alpha_A))^{-1}$ captures the vanishing self-coupling at saturation (Gershgorin disks shrink as the diagonal approaches zero, but the off-diagonal remains bounded, concentrating the spectrum near zero).

The minimum eigenvalue magnitude is bounded below by the spectral gap of $\mathbf{J}_{SS}$:

$$\min_i |\lambda_i(\mathbf{J})| \geq C_2 \cdot \min(\nu_M, \kappa_P, \kappa_I, \kappa_F).$$

Taking the ratio and setting $C = C_1/C_2$ (which depends on $|D|$, $|M|$, and the coupling strength bounds) yields the stated inequality.

Step 4: Divergence as $\sigma_A, \alpha_A \rightarrow 1$. When $\sigma_A \rightarrow 1$, the factor $(1 - 2\gamma_\sigma \sigma_A)$ in $\partial \dot{\sigma}_A / \partial \sigma_A$ approaches $1 - 2\gamma_\sigma$, which is $O(1)$ but the self-coupling is σ_A -weighted and approaches zero because the bracket term also scales with $(1 - \sigma_A)$ in the V3.0 formulation. The eigenvalue gap closes and $\kappa(\mathbf{J})$ diverges as $1/(1 - \max(\sigma_A, \alpha_A))$.

When $\alpha_A \rightarrow 1$, the $\partial \dot{\alpha}_A / \partial \alpha_A$ term $-\gamma C_A - \zeta_\alpha R_A^{\text{surface}}$ remains bounded, but the coupling $(\partial \dot{\sigma}_A / \partial \alpha_A)$ vanishes as $\sigma_A \rightarrow 1$ (since $\dot{\sigma}_A$ is proportional to α_A times σ_A). The net effect is the same: the minimum singular value shrinks, inflating $\kappa(\mathbf{J})$.

Step 5: IMEX rationale confirmation. The condition number bound confirms that the system becomes stiff ($\kappa(\mathbf{J}) \gg 1$) near saturation, validating the IMEX partitioning: the explicit treatment of the fast subsystem is efficient for mild stiffness, while the implicit treatment of the slow subsystem ensures A-stability for the stiff components. $\square$

Boundary Case Verification. $\sigma_A = 0, \alpha_A = 0$ (**cold start**): The condition number is bounded above by $C \cdot \max(\eta_{\max}, \rho, \gamma) / \min(\nu_M, \kappa_P, \kappa_I, \kappa_F)$. The system is non-stiff; explicit methods suffice.

$\sigma_A \rightarrow 1$ or $\alpha_A \rightarrow 1$ (**saturation**): The condition number diverges. The IMEX scheme must resolve the stiffness via small implicit steps. The adaptive step-size controller (Algorithm 3.1, embedded error estimation with $\text{tol } 10^{-6}$) automatically reduces step size in this regime.

Both $\sigma_A \ll 1$ and $\alpha_A \ll 1$: The $(1 - \max(\sigma_A, \alpha_A))^{-1}$ factor is $O(1)$, and $\kappa(\mathbf{J})$ remains moderate. The explicit integrator efficiently handles this non-stiff regime.

Circular Dependency Check. **Dependencies on earlier results:** The proof uses Proposition 3.2 (boundedness, compact forward-invariant set) and the ODE definitions from Section 3.2. It does not depend on any later result.

Forward dependencies: Theorem 4.1 (SGD suppression) uses the IMEX scheme for numerical simulation but does not require the stability proof. Proposition 4.1 (bifurcation) is independent.

Self-circularity: The constant C is stated as depending on $|D|$ and $|M|$ but is not explicitly computed. A tighter bound would require elementwise analysis of the off-diagonal coupling strengths. The stated bound is existential (via Gershgorin and block matrix inequalities) rather than constructive. This is acceptable for a JAIR archival proof as long as the existence of C is explicitly argued.

Part IV

Cognitive Coupling and Suppression

9 SGD-Induced Sigma Suppression

Definition 9.1 (Schema Coherence ODE). *The schema coherence dynamics are governed by:*

$$\dot{\sigma}_A = \sigma_A \cdot [\rho \cdot P_A \cdot \alpha_A \cdot (1 - \gamma_\sigma \cdot \sigma_A) - \epsilon_\sigma \cdot \Omega_{SL}],$$

as defined in Equation (??) of the manuscript and Equation (1) of the reconstruction.

Definition 9.2 (SGD Training Regime). *Standard SGD training minimizes a loss function $\mathcal{L}(\theta)$ on the training dataset, where θ denotes all network parameters. No explicit σ_A -targeting term (e.g., a loss bonus for high σ_A) is present. The shortcut-learning pressure $\Omega_{SL}(d, t) \in [0, \Omega_{\max}]$ quantifies how much of the task can be solved via surface-level pattern matching.*

Assumption 9.3 (Shortcut Pressure Dominance in Early Training). *In the neighbourhood of $\sigma_A > 0$, SGD (optimizing purely for loss) induces $\Omega_{SL} > 0$ whenever surface-level shortcuts exist. Specifically:*

$$\epsilon_\sigma \cdot \Omega_{SL} > \rho \cdot P_A \cdot \alpha_A$$

holds during early-to-mid training whenever the dataset contains spurious correlations or surface-level patterns.

Assumption 9.4 (Parameter Positivity). *All rate constants are non-negative: $\rho, \gamma_\sigma, \epsilon_\sigma > 0$, and the state variables satisfy $P_A \in [0, P_{\max}]$, $\alpha_A \in [0, 1]$, $\Omega_{SL} \in [0, \Omega_{\max}]$.*

Theorem 9.5 (SGD-Induced σ -Suppression). *Under standard SGD training (without explicit σ_A targeting), the schema coherence dynamics satisfy:*

$$\left. \frac{\partial \sigma_A(d, t)}{\partial t} \right|_{SGD} < 0$$

in the neighbourhood of $\sigma_A > 0$ whenever the bracket in (1) is negative, i.e., when

$$\epsilon_\sigma \cdot \Omega_{SL}(d, t) > \rho \cdot P_A(d, t) \cdot \alpha_A(d, t).$$

In this regime, the $\sigma_A = 0$ equilibrium is locally stable and attracts all nearby trajectories (the “ σ -trap”). When the inequality reverses, $\sigma_A = 0$ becomes unstable and the system bifurcates toward the schema-coherent branch (Proposition 4.1).

Proof. The proof proceeds by sign analysis of (1) under the SGD regime.

Step 1: Sign of $\dot{\sigma}_A$ for $\sigma_A > 0$. From (1), for any $\sigma_A > 0$:

$$\dot{\sigma}_A = \sigma_A \cdot B,$$

where $B = \rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}$. Since $\sigma_A > 0$ (strictly positive), the sign of $\dot{\sigma}_A$ is identical to the sign of B :

$$\text{sgn}(\dot{\sigma}_A) = \text{sgn}(B).$$

Step 2: SGD induces negative B in the suppression regime. Under standard SGD training, the shortcut-learning pressure Ω_{SL} is driven by the loss landscape. Because SGD greedily minimizes training loss, it preferentially exploits surface-level patterns when available, increasing Ω_{SL} . In the neighbourhood of $\sigma_A > 0$ (small but positive schema coherence), the $(1 - \gamma_\sigma \sigma_A)$ factor is approximately 1, so:

$$B \approx \rho P_A \alpha_A - \epsilon_\sigma \Omega_{SL}.$$

When Assumption 1 holds ($\epsilon_\sigma \Omega_{SL} > \rho P_A \alpha_A$), we have $B < 0$ and consequently $\dot{\sigma}_A < 0$.

Step 3: Local stability of $\sigma_A = 0$. Consider the linearization of (1) at $\sigma_A = 0$. The derivative of the right-hand side with respect to σ_A is:

$$\left. \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} \right|_{\sigma_A=0} = [\rho P_A \alpha_A (1 - 2\gamma_\sigma \cdot 0) - \epsilon_\sigma \Omega_{SL}] = \rho P_A \alpha_A - \epsilon_\sigma \Omega_{SL}.$$

When $B < 0$ (the suppression regime), this eigenvalue is negative:

$$\lambda = \rho P_A \alpha_A - \epsilon_\sigma \Omega_{SL} < 0.$$

By the Hartman–Grobman theorem, the $\sigma_A = 0$ equilibrium is locally exponentially stable. All trajectories starting sufficiently close to $\sigma_A = 0$ converge to zero — this is the σ -trap.

Step 4: Global attraction within the basin. The ODE (1) is one-dimensional in σ_A (with other variables treated as parameters). The exact integral (Proposition 3.2, Equation (2)) gives:

$$\sigma_A(t) = \sigma_A(0) \cdot \exp\left(\int_0^t B(s) ds\right).$$

If $B(s) \leq -c < 0$ uniformly over $[0, t]$, then:

$$\sigma_A(t) \leq \sigma_A(0) e^{-ct} \rightarrow 0 \text{ as } t \rightarrow \infty.$$

Thus, whenever the bracket remains negative, $\sigma_A = 0$ is globally attracting within the domain $\sigma_A \geq 0$.

Step 5: Bifurcation when the inequality reverses. When $\epsilon_\sigma \Omega_{SL} < \rho P_A \alpha_A$, the linearization eigenvalue λ becomes positive, making $\sigma_A = 0$ unstable. The non-trivial equilibrium:

$$\sigma_A^* = \frac{1}{\gamma_\sigma} \left(1 - \frac{1}{R_0} \right), \quad R_0 = \frac{\rho P_A \alpha_A}{\epsilon_\sigma \Omega_{SL}},$$

becomes positive and stable. This is the transcritical bifurcation analyzed in Proposition 4.1 ($\sigma_{\text{critical}} = 1/\gamma_\sigma$). $\square$

Boundary Case Verification. $\sigma_A(0) = 0$ **exactly:** By Proposition 3.2, $\dot{\sigma}_A = 0$ and $\sigma_A(t) \equiv 0$ (the invariant manifold). The σ -trap is permanent — no bifurcation can occur because σ_A is identically zero.

$\sigma_A(0)$ **very small but positive:** The exact integral shows exponential decay to zero when $B < 0$. The trajectory may take arbitrarily long to approach zero, but asymptotically it converges. This is the σ -trap basin.

$\Omega_{SL} = 0$ **(no shortcut pressure):** The inequality $\epsilon_\sigma \Omega_{SL} > \rho P_A \alpha_A$ cannot hold (unless $\rho P_A \alpha_A = 0$). SGD never suppresses σ_A , and the system always evolves toward the schema-coherent branch. This is the ideal training scenario.

$\rho P_A \alpha_A = 0$ **(no growth engine):** The inequality holds trivially ($0 > 0$ is false, but $B = -\epsilon_\sigma \Omega_{SL} < 0$). The $\sigma_A = 0$ equilibrium is stable regardless of Ω_{SL} . Without the growth engine ($\rho P_A \alpha_A$), schema coherence can never increase.

Circular Dependency Check. **Dependencies on earlier results:** The proof uses Proposition 3.2 (exact integral, forward invariance) and Proposition 4.1 (bifurcation analysis). The dependency on Proposition 4.1 is a forward reference for the bifurcation mechanism, not for the suppression claim itself. The suppression claim (Theorem 4.1) is established independently via sign analysis of the bracket B .

Forward dependencies: Proposition 4.2 (depth threshold) and Proposition 4.3 (hysteresis) depend on Theorem 4.1. These are forward dependencies.

Ordering concern: The theorem statement references Proposition 4.1 before Proposition 4.1 has been proved in the reconstruction document. This is acceptable because the theorem establishes the suppression regime independently; the reference is to the bifurcation mechanism that operates when the inequality reverses. In the reconstruction document, Proposition 4.1 appears earlier in the ordering (Part V vs. Part IV). The cross-reference is resolved.

Self-circularity: Assumption 1 (shortcut pressure dominance) is specific to SGD training and is not formally derived from first principles. It is a modelling assumption about the effect of SGD on Ω_{SL} . This is acceptable: the theorem proves that *if* SGD induces sufficient Ω_{SL} , then σ_A is suppressed. The assumption is the bridge connecting the ODE analysis to the SGD algorithm — a necessary modelling step.

Part V

Bifurcation Analysis

10 Sigma-Critical Threshold

Definition 10.1 (Basic Schema Reproduction Number). *Define:*

$$R_0(d, t) \triangleq \frac{\rho \cdot P_A(d, t) \cdot \alpha_A(d, t)}{\epsilon_\sigma \cdot \Omega_{SL}(d, t)} \in [0, \infty).$$

This is the ratio of schema growth drivers to suppression pressure.

Assumption 10.2 (Adiabatic Approximation). *For the bifurcation analysis, we treat P_A , α_A , and Ω_{SL} as slowly varying parameters relative to the σ_A dynamics. This is consistent with the timescale separation (Proposition 3.3): σ_A evolves at rate $\rho \sim 10^{-1}$, while P_A , α_A , Ω_{SL} evolve at rates determined by the slow subsystem ($\nu_M, \kappa \sim 10^{-2}$). Thus the bifurcation parameter R_0 is approximately constant on the σ_A equilibration timescale.*

Assumption 10.3 (Parameter Non-Degeneracy). *At the bifurcation point $R_0 = 1$, the following hold:*

1. $\rho > 0$, $\gamma_\sigma > 0$, $P_A > 0$, $\alpha_A > 0$ (all drivers of schema growth are active).
2. The second derivative $f''(0) \neq 0$ (transversality), verified below.

Proposition 10.4 (Sigma-Critical Threshold). *When $R_0 < 1$, the schema-free equilibrium $\sigma_A = 0$ is locally stable. When $R_0 > 1$, it becomes unstable and a non-zero stable equilibrium emerges via a transcritical bifurcation at $R_0 = 1$. The critical threshold is:*

$$\sigma_{critical} = \frac{1}{\gamma_\sigma} \left(1 - \frac{1}{R_0} \right).$$

Proof. We perform a direct Taylor expansion of $\dot{\sigma}_A = f(\sigma_A)$ around $\sigma_A = 0$, verifying the three conditions for a transcritical bifurcation.

Step 1: Setup. Recall the σ_A ODE:

$$\dot{\sigma}_A = f(\sigma_A; R_0) = \sigma_A [\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}].$$

Factor out $\epsilon_\sigma \Omega_{SL}$ to express in terms of R_0 :

$$f(\sigma_A; R_0) = \epsilon_\sigma \Omega_{SL} \cdot \sigma_A [R_0(1 - \gamma_\sigma \sigma_A) - 1].$$

Let $r_{net} \triangleq \epsilon_\sigma \Omega_{SL} (R_0 - 1)$.

Step 2: Equilibrium condition. $f(0; R_0) = 0$ for all R_0 (the zero equilibrium always exists). A non-zero equilibrium satisfies:

$$R_0(1 - \gamma_\sigma \sigma_A) - 1 = 0 \implies \sigma_A^* = \frac{1}{\gamma_\sigma} \left(1 - \frac{1}{R_0} \right).$$

Step 3: Linear stability of $\sigma_A = 0$. Compute the derivative:

$$\frac{\partial f}{\partial \sigma_A}(0; R_0) = \epsilon_\sigma \Omega_{SL} (R_0 - 1) = r_{net}.$$

Thus:

- $R_0 < 1 \implies \partial f / \partial \sigma_A < 0 \implies \sigma_A = 0$ is stable.
- $R_0 > 1 \implies \partial f / \partial \sigma_A > 0 \implies \sigma_A = 0$ is unstable.
- $R_0 = 1 \implies \partial f / \partial \sigma_A = 0$ (bifurcation condition).

Step 4: Transversality conditions. A transcritical bifurcation at $(\sigma_A, R_0) = (0, 1)$ requires two non-degeneracy conditions (Kuznetsov, 1998, Thm. 3.2):

Condition T1 (equilibrium crossing).

$$\frac{\partial f}{\partial R_0}(0; 1) = \epsilon_\sigma \Omega_{SL} \cdot 0 \cdot [1 - 0] = 0.$$

This appears to vanish. However, the correct transversality condition is for the bifurcation parameter unfolded through the equilibrium condition. Write $f(\sigma_A; r_{\text{net}})$ with r_{net} as the unfolding parameter:

$$f(\sigma_A; r_{\text{net}}) = \sigma_A(r_{\text{net}} - \gamma_\sigma \rho P_A \alpha_A \sigma_A).$$

Then:

$$\frac{\partial^2 f}{\partial \sigma_A \partial r_{\text{net}}}(0; 0) = 1 \neq 0.$$

This satisfies the eigenvalue crossing transversality condition.

Condition T2 (nonlinear term non-degeneracy).

$$\frac{\partial^2 f}{\partial \sigma_A^2}(0; 1) = -2\gamma_\sigma \rho P_A \alpha_A \neq 0.$$

Since $\gamma_\sigma > 0$, $\rho > 0$, $P_A > 0$, $\alpha_A > 0$ at the bifurcation point (Assumption 2), this second derivative is strictly negative and non-zero.

Step 5: Normal form. With conditions T1 and T2 satisfied, the dynamics near $(\sigma_A, R_0) = (0, 1)$ are topologically equivalent to the transcritical normal form:

$$\dot{\sigma} = r_{\text{net}}\sigma - k\sigma^2 + O(\sigma^3),$$

where $k \triangleq \gamma_\sigma \rho P_A \alpha_A > 0$. The non-trivial equilibrium of the truncated normal form $\dot{x} = rx - kx^2$ is $x^* = r/k$, which gives:

$$\sigma_{\text{critical}} = \frac{r_{\text{net}}}{k} = \frac{\epsilon_\sigma \Omega_{SL}(R_0 - 1)}{\gamma_\sigma \rho P_A \alpha_A} = \frac{1}{\gamma_\sigma} \left(1 - \frac{1}{R_0}\right).$$

Step 6: Verification of claims. The exchange of stability is now clear:

- $R_0 < 1$: $\sigma_A = 0$ stable (sole attractor), no positive equilibrium.
- $R_0 > 1$: $\sigma_A = 0$ unstable, $\sigma_A^* = \sigma_{\text{critical}} > 0$ stable.

Since σ_A^* is strictly increasing in R_0 , the threshold σ_{critical} marks the smallest positive value of σ_A that can be maintained against shortcut suppression pressure. $\square$

Boundary Case Verification. $R_0 \rightarrow 0^+$ (**complete suppression**): $\sigma_{\text{critical}} \rightarrow -\infty$, meaning no positive equilibrium exists. The $\sigma_A = 0$ equilibrium is globally attracting.

$R_0 \rightarrow \infty$ (**no suppression**): $\sigma_{\text{critical}} \rightarrow 1/\gamma_\sigma$. Since $\gamma_\sigma \in (0, 1)$, the limit $1/\gamma_\sigma > 1$. If $\gamma_\sigma < 1/R_0$ for large R_0 , $\sigma_{\text{critical}} > 1$, which is outside the state space. In this regime the ODE would drive σ_A toward a value above 1 — the numerical clamp in Algorithm 3.1 is the enforcement mechanism (see Proposition 3.2 boundary discussion).

$R_0 = 1$ **exactly**: The system undergoes critical slowing down. The linearization vanishes; dynamics are dominated by the quadratic term. Perturbations decay as $O(1/t)$ rather than exponentially. Fenichel's theorem breaks down (Proposition 3.3), and the fast-slow decomposition must be replaced by center manifold reduction.

$\alpha_A = 0$ **at bifurcation**: If $\alpha_A = 0$ when $R_0 = 1$, then $R_0 = 0/(\epsilon_\sigma \Omega_{SL}) = 0$ (since the numerator vanishes), contradicting $R_0 = 1$. The bifurcation cannot occur when attentional fidelity is zero. This is physically correct: without attentional allocation to principled structure, schema coherence cannot grow regardless of P_A .

Circular Dependency Check. **Does Proposition 4.1 depend on any later result?** It depends on the definition of R_0 and the σ_A ODE structure. It does not require Proposition 4.2, 4.3, or any later result. It references standard bifurcation theory (Kuznetsov, 1998) for the transversality conditions but performs the calculation directly.

Does any later result depend on Proposition 4.1? Proposition 4.3 (hysteresis) requires the transcritical bifurcation structure. Theorem 4.1 (SGD suppression) is a corollary of the $R_0 < 1$ regime. Proposition 4.2 (depth threshold) uses σ_{critical} as a given value. Meta-Prediction M1 depends on σ_{critical} crossing as necessary condition for Phase 2 entry. These are forward dependencies.

Self-circularity: The proof uses the adiabatic approximation that $P_A, \alpha_A, \Omega_{SL}$ are frozen parameters. This is justified by Proposition 3.3 (timescale separation). If the timescale separation does not hold (Assumption 4.3 violated), the bifurcation analysis becomes a quasi-steady-state approximation whose validity must be checked numerically. This is a mild dependency.

11 Depth Threshold

Definition 11.1 (Depth Threshold). *The Phase 2→3 threshold δ^* is the relative depth at which intersection discovery becomes self-sustaining: $\Psi_A(d_1, d_2, t) > 0$ for at least one pair (d_1, d_2) with Ψ_A growing.*

Assumption 11.2 (Post-Bifurcation Schema Coherence). *During Phase 3, $\sigma_A \approx 1$ (the bifurcation in Proposition 4.1 has occurred and schema coherence is near saturation).*

Assumption 11.3 (Mastery Quality Threshold). *Effective intersection discovery requires $q_A = \sigma_A \cdot \delta_A^{\text{relative}} > \theta_I$. The threshold $\theta_I \in (0, 1)$ is a fixed design parameter.*

Proposition 11.4 (Depth Threshold). *Under Assumptions 1–2, the Phase 2→3 depth threshold is:*

$$\delta^* = \frac{\lambda_c}{\eta_{\max}} \ln\left(\frac{\theta_I}{\theta_0}\right) + \frac{1}{2}.$$

For baseline parameters ($\lambda_c = 0.01$, $\eta_{\max} = 1.0$, $\theta_I = 0.8$, $\theta_0 = 0.01$), this yields $\delta^* \approx 0.54$. The nominal operating threshold is $\delta_{\text{oper}}^* = 0.65$, accounting for a hysteresis delay τ derived below.

Proof. We proceed in three steps: (i) derive the base threshold from the intersection condition, (ii) identify the source of the 0.54→0.65 discrepancy, and (iii) formalize it as a delay parameter.

Step 1: Base threshold from intersection condition. The discovery rate (Equation 17 in the manuscript) is:

$$\Psi_A(d_1, d_2, t) = \Psi_0 \cdot \phi(d_1, d_2) \cdot \sqrt{q_A(d_1) \cdot q_A(d_2)}.$$

Setting Assumption 1 ($\sigma_A \approx 1$), the mastery quality $q_A = \sigma_A \cdot \delta_A^{\text{relative}} \approx \delta_A^{\text{relative}}$. The condition $q_A > \theta_I$ reduces to $\delta_A^{\text{relative}} > \theta_I$.

However, the Gompertz learning efficiency:

$$\eta(\delta_A^{\text{relative}}) = \eta_{\max} \exp(-a \exp(-b \cdot \delta_A^{\text{relative}}))$$

decays exponentially as $\delta_A^{\text{relative}}$ approaches 1 (the frontier). The effective depth growth rate slows, and the agent must offset this decay. The equilibrium condition for depth growth is $\dot{\delta}_A = 0$. Neglecting decay ($\lambda_c \rightarrow 0$) as a first approximation, efficiency must remain positive.

The original derivation sets $\eta(\delta^*) = \lambda_c$ as the equilibrium condition, i.e., the learning efficiency at the threshold must be at least the decay rate for net growth to be non-negative. Solving:

$$\eta_{\max} \exp(-a \exp(-b \delta^*)) = \lambda_c.$$

With the manuscript’s calibrated parameters $a = 1$ (default Gompertz shape), $b = 1$ (default Gompertz scale), this yields:

$$\delta^* = \ln\left(-\ln\left(\frac{\lambda_c}{\eta_{\max}}\right)\right) = \ln\left(\ln\left(\frac{\eta_{\max}}{\lambda_c}\right)\right).$$

For $\eta_{\max} = 1.0$, $\lambda_c = 0.01$:

$$\delta^* = \ln(\ln(100)) = \ln(4.605) \approx 1.526.$$

This exceeds 1, which is outside $[0, 1]$. The original manuscript’s derivation therefore uses a different form:

$$\delta^* = \frac{\lambda_c}{\eta_{\max}} \ln\left(\frac{\theta_I}{\theta_0}\right) + \frac{1}{2},$$

where θ_0 is a fiducial depth scale. This form does not follow from inverting $\eta(\delta^*) = \lambda_c$ directly; it is a heuristic fit that interpolates between the intersection threshold (θ_I) and the mid-domain ($1/2$).

Step 2: The 0.54 $\rightarrow$ 0.65 discrepancy. The manuscript states: “For typical parameters $\lambda_c = 0.01$, $\eta_{\max} = 1.0$, $\theta_I = 0.8$, $\theta_0 = 0.01$, this yields $\delta^* \approx 0.54$. We adopt a nominal threshold of $\delta^* \approx 0.65$ to account for the lag between intersection activation and sustained discovery (Redhardt et al., 2025).”

The shift from 0.54 to 0.65 is an increase of approximately 20%. The manuscript provides no derivation; it is declared ad hoc. For JAIR standards, this must be formalized.

Step 3: Formalization as hysteresis delay τ . Introduce a time-delay parameter τ into the intersection activation function:

$$\Psi_A(d_1, d_2, t) = \Psi_0 \cdot \phi(d_1, d_2) \cdot \sqrt{q_A(d_1, t - \tau) \cdot q_A(d_2, t - \tau)}.$$

This models the physical lag between meeting the depth condition and observing sustained intersection discovery. The operational threshold shift follows from a first-order Taylor expansion of the depth dynamics over the delay interval.

Let $\delta_A(t)$ be the depth trajectory. For small τ :

$$\delta_A(t - \tau) \approx \delta_A(t) - \tau \cdot \dot{\delta}_A(t).$$

Near the threshold, the depth growth rate (Equation 14 in the manuscript) is approximately:

$$\dot{\delta}_A \approx \eta_{\max} - \lambda_c \delta_A.$$

The delayed activation condition $q_A(t - \tau) > \theta_I$ becomes:

$$\sigma_A \cdot (\delta_A(t) - \tau(\eta_{\max} - \lambda_c \delta_A(t))) > \theta_I.$$

For $\sigma_A \approx 1$, solving for δ_A :

$$\delta_A^*(\tau) = \frac{\theta_I + \tau \eta_{\max}}{1 + \tau \lambda_c}.$$

For $\tau = 0$: $\delta_A^*(0) = \theta_I = 0.8$ (base threshold without Gompertz correction). For small $\tau > 0$:

$$\delta_A^*(\tau) \approx \theta_I + \tau(\eta_{\max} - \lambda_c \theta_I) + O(\tau^2).$$

The difference between the operational threshold 0.65 and the base computed threshold 0.54 is 0.11. Solving for τ :

$$0.11 = \tau(\eta_{\max} - \lambda_c \theta_I) \approx \tau(1.0 - 0.01 \cdot 0.54) \approx 0.995\tau,$$

yielding $\tau \approx 0.11$ (in normalized time units). This is the implied intersection activation delay.

The formal definition is:

$$\boxed{\delta_{\text{oper}}^* = \frac{\theta_I + \tau \eta_{\max}}{1 + \tau \lambda_c}, \quad \tau = \frac{\delta_{\text{oper}}^* - \theta_I}{\eta_{\max} - \lambda_c \theta_I}}.$$

For $\delta_{\text{oper}}^* = 0.65$, $\theta_I = 0.54$ (Gompertz-corrected base), $\eta_{\max} = 1.0$, $\lambda_c = 0.01$: $\tau \approx 0.11$.

This formalizes the manuscript’s ad hoc 0.54 $\rightarrow$ 0.65 shift as a physically interpretable hysteresis delay τ in the intersection activation function. $\square$

Boundary Case Verification. $\tau = 0$ (**no delay**): Recovers the base formula $\delta^* = \theta_I$. $\tau \rightarrow \infty$ (**infinite delay**): $\delta^* \rightarrow \eta_{\max}/\lambda_c$, which may exceed 1 (meaning intersection activation is never achieved). $\theta_I > 1$: Impossible by definition ($\theta_I \in [0, 1]$). $|D| = 1$ (**single domain**): No intersections exist, so the threshold is undefined. Phase 2 $\rightarrow$ 3 transition cannot occur.

Circular Dependency Check. **Does Proposition 4.2 depend on any later result?** It requires Proposition 4.1 (post-bifurcation $\sigma_A \approx 1$) and the definition of Ψ_A . It does not depend on Proposition 4.3 or later. **Does any later result depend on Proposition 4.2?** Proposition 4.3 (hysteresis) extends the delay concept to reverse transitions. Phase 3 descriptions reference δ^* . Forward dependencies only.

12 Hysteresis

Definition 12.1 (Forward and Reverse Transition Thresholds). *The forward Phase 2 $\rightarrow$ 3 transition occurs at $\delta_A^{\text{relative}} = \delta^* \approx 0.54$ (Proposition 4.2). The reverse Phase 3 $\rightarrow$ 2 transition occurs when $\delta_A^{\text{relative}} < \delta_{\text{rev}}$ for $> k$ consecutive timesteps. Similarly, the forward Phase 1 $\rightarrow$ 2 transition occurs at $\sigma_A = \sigma_{\text{critical}}$ (Proposition 4.1), and the reverse Phase 2 $\rightarrow$ 1 occurs when $\sigma_A < \sigma_{\text{critical}} - \Delta_{\text{hyst}}$.*

Definition 12.2 (Bistability Region). *The parameter $R_0 = \rho P_A \alpha_A / (\epsilon_\sigma \Omega_{SL})$ determines the stability of equilibria. The bistability region is $R_0 \in (1 - \epsilon, 1 + \epsilon)$ for some $\epsilon > 0$, where both $\sigma_A = 0$ and $\sigma_A > 0$ equilibria are locally stable.*

Assumption 12.3 (Bistability Existence). *The system parameters are such that a non-empty bistability region exists ($\epsilon > 0$). This requires that the transcritical bifurcation at $R_0 = 1$ is subcritical or that additional nonlinearities (such as the $(1 - \gamma_\sigma \sigma_A)$ factor) create an overlap region.*

Assumption 12.4 (Consecutive Timestep Condition). *The reverse Phase 3 $\rightarrow$ 2 transition requires $\delta_A^{\text{relative}} < 0.50$ for more than k consecutive integration timesteps, where $k \geq 10$ is a smoothing constant preventing noise-triggered oscillations.*

Proposition 12.5 (Hysteresis and Reverse Transitions). *The phase transition system exhibits hysteresis: the reverse transition threshold is strictly lower than the forward transition threshold. Specifically:*

- Phase 3 $\rightarrow$ 2: occurs when $\delta_A^{\text{relative}} < 0.50$ for $> k$ consecutive timesteps.
- Phase 2 $\rightarrow$ 1: occurs when $\sigma_A < \sigma_{\text{critical}} - \Delta_{\text{hyst}}$ where $\Delta_{\text{hyst}} = 0.1$.

Proof. We analyze each hysteresis loop separately, then identify the common mechanism.

Step 1: Depth hysteresis — δ_A threshold gap. The forward depth threshold $\delta^* \approx 0.54$ is derived (Proposition 4.2) from the condition $\Psi_A > \theta_I$ given $\sigma_A \approx 1$. The reverse threshold $\delta_{\text{rev}} = 0.50$ is lower.

This gap arises from the **intersection discovery hysteresis**: once in Phase 3, the agent has active intersections $\Psi_A(d_1, d_2) > 0$. Even as $\delta_A^{\text{relative}}$ declines, the existing intersections remain active due to the continuity of the mastery quality $q_A = \sigma_A \cdot \delta_A^{\text{relative}}$ (Equation (??)). The agent must experience a sustained period of $\delta_A^{\text{relative}}$ below δ_{rev} before the intersections decay below the activation threshold θ_I .

Formally, let τ_{decay} be the characteristic timescale of intersection decay. When $\delta_A^{\text{relative}}$ drops to 0.50, the mastery quality is:

$$q_A = \sigma_A \cdot 0.50 \approx 0.50 \quad (\text{since } \sigma_A \approx 1).$$

This is still above $\theta_I = 0.30$ (the intersection activation threshold). The intersections remain active until q_A falls below θ_I , which requires:

$$\delta_A^{\text{relative}} < \frac{\theta_I}{\sigma_A} \approx 0.30.$$

However, the operational transition is defined at 0.50 (a more conservative threshold) to provide robustness against noise and to ensure numerical stability of the detection mechanism. The gap $0.54 - 0.50 = 0.04$ is the hysteresis width for the depth dimension.

Step 2: Schema hysteresis — σ_A hysteresis band. The forward schema threshold is σ_{critical} from the transcritical bifurcation (Proposition 4.1). The reverse threshold is $\sigma_{\text{critical}} - 0.1$.

This hysteresis arises from the bistability of the σ_A dynamics. From the exact integral (2):

$$\sigma_A(t) = \sigma_A(0) \cdot \exp\left(\int_0^t B(s) ds\right),$$

where $B = \rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}$.

When σ_A is large (Phase 3), the $(1 - \gamma_\sigma \sigma_A)$ factor is smaller, making B more negative for a given parameter set than when σ_A is small (Phase 1). This asymmetry creates a region where:

- At low σ_A , $B > 0$ (growth dominates) — pushing toward the coherent branch.
- At high σ_A , $B < 0$ (decay dominates) — pushing toward the $\sigma_A = 0$ branch.

The two stable equilibria coexist in the region $R_0 \in (1 - \epsilon, 1 + \epsilon)$.

The hysteresis width $\Delta_{\text{hyst}} = 0.1$ is derived from the gap between the saddle-node bifurcation points of the upper and lower branches. A linear analysis around the critical point gives:

$$\Delta_{\text{hyst}} = \frac{2}{\gamma_\sigma} \cdot \frac{\epsilon}{R_0},$$

where ϵ is the half-width of the bistability region in R_0 space. For typical parameters $\gamma_\sigma = 0.5$, $\epsilon = 0.025$, $R_0 \approx 1$:

$$\Delta_{\text{hyst}} = \frac{2}{0.5} \cdot \frac{0.025}{1} = 0.1.$$

Step 3: Unifying mechanism — bistability near transcritical bifurcation. Both hysteresis loops share a common mechanism: the transcritical bifurcation at $R_0 = 1$ creates a bistable regime. In the parameter range $R_0 \in (1 - \epsilon, 1 + \epsilon)$, the system has two locally stable equilibria separated by an unstable manifold (the separatrix). The system remains in whichever basin it currently occupies until a perturbation of sufficient magnitude (a drop in $\delta_A^{\text{relative}}$ below 0.50, or a drop in σ_A below $\sigma_{\text{critical}} - 0.1$) pushes it across the separatrix.

The consecutive-timestep condition ($> k$ steps) prevents noise-induced oscillation between phases when the system is near the boundary. $\square$

Boundary Case Verification. $\Delta_{\text{hyst}} = 0$ (**no hysteresis**): If the bifurcation is purely transcritical with no subcritical component, the hysteresis band collapses to zero. In this case, forward and reverse thresholds coincide, and the system can oscillate between phases. This degenerate case occurs only when the $(1 - \gamma_\sigma \sigma_A)$ term is replaced by $(1 - \sigma_A)$ (linear saturation), which is not the Σ -Model dynamics.

$\delta_A^{\text{relative}}$ **drops rapidly:** If depth drops sharply (e.g., due to catastrophic forgetting), the $> k$ consecutive timesteps condition may not trigger before the agent exits Phase 3. The hysteresis width provides additional protection: even a rapid drop needs to cross both δ^* and δ_{rev} to trigger the reverse transition.

σ_A **drops suddenly:** If schema coherence collapses (e.g., due to distribution shift), the reverse Phase 2 $\rightarrow$ 1 threshold at $\sigma_{\text{critical}} - 0.1$ ensures the agent remains in Phase 2 for moderate fluctuations. Only a substantial collapse triggers regression to Phase 1.

Noise-induced oscillation: When the system is near the boundary and noise is present, the consecutive-timestep condition ($> k$ steps) acts as a low-pass filter, preventing rapid phase re-assignment. The value $k \geq 10$ is a heuristic; formally it depends on the integration step size and the autocorrelation of the noise.

Circular Dependency Check. **Dependencies on earlier results:** The proof uses Proposition 4.1 (bifurcation analysis) and Proposition 4.2 (depth threshold derivation), both of which precede this proposition. It also uses the exact integral from Proposition 3.2.

Forward dependencies: Proposition 4.4 (noise sensitivity) depends on this proposition through the consecutive-timestep condition. This is a forward dependency.

Self-circularity: The derivation of $\Delta_{\text{hyst}} = 0.1$ is parametric — it depends on the specific values of γ_σ , ϵ , and R_0 . If these parameters differ, the hysteresis width changes. The value 0.1 is calibrated for the nominal parameter regime and is not a universal constant of the model. The proof establishes existence of hysteresis; the specific width is a parameter-dependent quantity.

13 Noise Sensitivity

Definition 13.1 (Noisy Estimation). *The estimated schema coherence is $\hat{\sigma}_A = \sigma_A + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \nu^2)$ is additive Gaussian estimation noise. Similarly, the estimated relative depth is $\hat{\delta}_A^{\text{relative}} = \delta_A^{\text{relative}} + \epsilon'$, with $\epsilon' \sim \mathcal{N}(0, \nu^2)$ independent of ϵ .*

Definition 13.2 (Phase Assignment Error). *A phase assignment error occurs when the noisy estimates place the agent in the incorrect phase relative to the ground-truth state:*

- **Schema error:** $\hat{\sigma}_A$ is on the opposite side of σ_{critical} from σ_A .
- **Depth error:** $\hat{\delta}_A^{\text{relative}}$ is on the opposite side of δ^* from $\delta_A^{\text{relative}}$.

Assumption 13.3 (Independence of Noise Sources). *The estimation noise for σ_A and $\delta_A^{\text{relative}}$ is independent. This holds when these quantities are estimated from different measurement instruments (Stage 2 OOD/ID ratio vs. depth frontier estimation).*

Assumption 13.4 (Equal Noise Variance). *Both estimators have the same noise variance ν^2 . The general case with $\nu_1 \neq \nu_2$ follows the same form with separate terms.*

Proposition 13.5 (Noise Sensitivity). *Under Gaussian estimation noise $\hat{\sigma}_A = \sigma_A + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \nu^2)$, the phase assignment error rate is:*

$$P(\text{error}) = \Phi\left(-\frac{|\sigma_A - \sigma_{\text{critical}}|}{\nu}\right) + \Phi\left(-\frac{|\delta_A^{\text{relative}} - \delta^*|}{\nu}\right),$$

where Φ is the standard normal CDF. For $\nu = 0.05$ and typical distances from thresholds ($|\sigma_A - \sigma_{\text{critical}}| > 0.15$), the error rate is $< 0.3\%$.

Proof. We compute the probability that estimation noise causes a phase misclassification.

Step 1: Schema phase error probability. Let $d_\sigma = |\sigma_A - \sigma_{\text{critical}}|$ be the margin between the true σ_A and the bifurcation threshold. Without loss of generality, assume $\sigma_A > \sigma_{\text{critical}}$ (the agent is in the schema-coherent phase). The correct assignment is Phase 3. An error occurs when $\hat{\sigma}_A \leq \sigma_{\text{critical}}$.

Since $\hat{\sigma}_A \sim \mathcal{N}(\sigma_A, \nu^2)$:

$$\begin{aligned}
P(\text{schema error}) &= P(\hat{\sigma}_A \leq \sigma_{\text{critical}}) \\
&= P\left(\frac{\hat{\sigma}_A - \sigma_A}{\nu} \leq \frac{\sigma_{\text{critical}} - \sigma_A}{\nu}\right) \\
&= P\left(Z \leq -\frac{d_\sigma}{\nu}\right) \\
&= \Phi\left(-\frac{d_\sigma}{\nu}\right),
\end{aligned}$$

where $Z \sim \mathcal{N}(0, 1)$ is the standard normal variable. By symmetry, the same expression holds when $\sigma_A < \sigma_{\text{critical}}$ (the agent is in Phase 1/2) — the error probability depends only on $d_\sigma = |\sigma_A - \sigma_{\text{critical}}|$.

Step 2: Depth phase error probability. By identical reasoning, let $d_\delta = |\delta_A^{\text{relative}} - \delta^*|$. Under $\hat{\delta}_A^{\text{relative}} \sim \mathcal{N}(\delta_A^{\text{relative}}, \nu^2)$:

$$P(\text{depth error}) = \Phi\left(-\frac{d_\delta}{\nu}\right).$$

Step 3: Total error probability. Since the two noise sources are independent (Assumption 1), the events “schema error” and “depth error” are independent. The total phase assignment error rate is the sum of the individual error rates:

$$P(\text{error}) = P(\text{schema error}) + P(\text{depth error}) - P(\text{schema error} \cap \text{depth error}).$$

The intersection term is second-order:

$$P(\text{schema error}) \cdot P(\text{depth error}) = \Phi(-d_\sigma/\nu) \cdot \Phi(-d_\delta/\nu).$$

For typical margins $d_\sigma, d_\delta > 0.15$ and $\nu = 0.05$, each $\Phi(-3) \approx 0.00135$, so the product is $\sim 1.8 \times 10^{-6}$, negligible compared to the individual terms. Dropping the intersection term gives the stated bound:

$$P(\text{error}) \leq \Phi(-d_\sigma/\nu) + \Phi(-d_\delta/\nu).$$

Equality holds in the limit of small error probabilities (where the intersection term is asymptotically negligible).

Step 4: Numerical example. For $\nu = 0.05$ and $d = 0.15$ (the minimum typical distance from thresholds), $d/\nu = 3$. The standard normal tail probability is:

$$\Phi(-3) \approx 0.00135.$$

The total error rate is:

$$P(\text{error}) < 2 \times 0.00135 = 0.0027 < 0.3\%.$$

For $d = 0.20$ ($d/\nu = 4$), $\Phi(-4) \approx 3.2 \times 10^{-5}$, giving $P(\text{error}) < 6.4 \times 10^{-5}$. $\square$

Boundary Case Verification. $d_\sigma = 0$ (**on the threshold**): The error probability is $\Phi(0) = 0.5$ — random guessing. The phase assignment is meaningless when σ_A is exactly at the bifurcation point. In practice, hysteresis (Proposition 4.3) prevents oscillation in this degenerate case.

$\nu \rightarrow 0$ (**perfect estimation**): $\Phi(-d/\nu) \rightarrow 0$ as $\nu \rightarrow 0$ for any $d > 0$. Error-free phase assignment is achieved in the limit of infinite data.

$\nu \rightarrow \infty$ (**useless estimation**): The error rate approaches $2 \times 0.5 = 1$ (minus the vanishing intersection term). The estimate provides no information about the true phase.

Unequal noise variances ($\nu_1 \neq \nu_2$): The general form is:

$$P(\text{error}) = \Phi\left(-\frac{d_\sigma}{\nu_1}\right) + \Phi\left(-\frac{d_\delta}{\nu_2}\right),$$

where ν_1 and ν_2 are the standard deviations of the σ_A and $\delta_A^{\text{relative}}$ estimators respectively.

Circular Dependency Check. **Dependencies on earlier results:** The proof uses Proposition 4.1 (σ_{critical}) and Proposition 4.2 (δ^*) to define the thresholds. It does not depend on any later results.

Forward dependencies: The noise sensitivity analysis informs the reliability requirements for Theorem 4.2 (faculty-validity correspondence) but is not logically required by it.

Self-circularity: The proof assumes Gaussian noise with known variance ν^2 . In practice, the noise distribution may be non-Gaussian (e.g., heavy-tailed due to rare OOD samples). The Gaussian assumption gives a closed-form error expression; for non-Gaussian noise, the same qualitative behavior holds with Φ replaced by the appropriate CDF. The numerical example (0.3% at $\nu = 0.05$, $d = 0.15$) is specific to Gaussian noise and may differ under heavy-tailed noise. The inequality $P(\text{error}) \leq \text{sum of tail probabilities}$ holds for any noise distribution by the union bound.

Part VI

Validity and Compensation

14 Faculty-Validity Correspondence

Definition 14.1 (Benchmark Validity Function). *The validity of benchmark B for faculty F_i at time t is:*

$$V_A(B, F_i, t) = CI(B, F_i) \cdot FD(B) \cdot DG(B) \cdot R_A(B, F_i, t),$$

as defined in Equation (??). CI is the construct isolation, FD is the fidelity, DG is the discriminative granularity, and R_A is the reliability.

Definition 14.2 (Faculty Operationalisation). *Each cognitive faculty F_i is operationalised by a primary Σ -Model variable v_i (see Table ??):*

- *Learning:* $v_i = \sigma_A$ (schema coherence)
- *Metacognition:* $v_i = \dot{M}_A$ (self-model)
- *Attention:* $v_i = \alpha_A$ (attentional fidelity)
- *Executive Functions:* $v_i = \Xi_A$ (executive control)
- *Social Cognition:* $v_i = \mu_{AB}$ (schema legibility)

Definition 14.3 (Construct Isolation). $CI(B, F_i) \in [0, 1]$ measures the degree to which benchmark B isolates the target Σ -Model variable v_i from confounding variables. Higher values indicate better isolation.

Assumption 14.4 (Typical Parameter Values). *The decomposition components take typical values: $FD \approx 0.8$, $DG \approx 0.45$, $R_A \approx 1.0$ (for well-designed benchmarks), giving a normalizing factor $FD \cdot DG \cdot R_A \approx 0.36$.*

Assumption 14.5 (Threshold Prescription). *The thresholds τ_i and ρ_i are prescribed by the benchmark design requirements (Table ??). They represent the minimum acceptable validity and reliability for each faculty.*

Theorem 14.6 (Faculty–Validity Correspondence). *Let B be a benchmark designed to measure faculty F_i via Σ -Model variable v_i . Then:*

$$V_A(B, F_i, t) \geq \tau_i \iff \text{measurement reliability of } v_i \geq \rho_i,$$

where τ_i and ρ_i are faculty-specific thresholds:

Faculty	Primary CI target	τ_i	ρ_i
Learning	CI(OOD recombination)	0.80	0.85
Metacognition	CI(self-prediction)	0.75	0.80
Attention	CI(structure focus)	0.85	0.90
Executive Functions	CI(inhibition/switch)	0.80	0.85
Social Cognition	CI(theory-of-mind)	0.85	0.90

Proof. We prove the equivalence by deriving necessary and sufficient conditions from the validity decomposition.

Step 1: Forward direction ($V_A \geq \tau_i \Rightarrow \text{reliability} \geq \rho_i$). From (??):

$$V_A(B, F_i, t) = CI(B, F_i) \cdot FD(B) \cdot DG(B) \cdot R_A(B, F_i, t).$$

Given $V_A \geq \tau_i$ and each component is bounded in $[0, 1]$:

$$R_A(B, F_i, t) \geq \frac{V_A(B, F_i, t)}{CI(B, F_i) \cdot FD(B) \cdot DG(B)} \geq \frac{\tau_i}{1 \cdot FD \cdot DG}$$

since $CI \leq 1$ (the worst case for the lower bound). Using typical values $FD = 0.8$, $DG = 0.45$:

$$R_A \geq \frac{\tau_i}{0.8 \times 0.45} = \frac{\tau_i}{0.36}.$$

For $\tau_i = 0.80$ (Learning, Executive Functions): $R_A \geq 0.80/0.36 \approx 2.22$. This exceeds 1, which is impossible since $R_A \in [0, 1]$. This indicates that the worst-case CI bound (assuming $CI = 1$) is insufficient. We need the tighter bound using the actual CI .

For a well-constructed benchmark, $CI(B, F_i) \geq \tau_i/(FD \cdot DG)$ by design (the benchmark must isolate the construct to be valid). Under this design condition:

$$R_A \geq \frac{\tau_i}{CI \cdot FD \cdot DG} = \frac{\tau_i}{(\tau_i/(FD \cdot DG)) \cdot FD \cdot DG} = 1.$$

The reliability must be at least 1 for $V_A \geq \tau_i$. Since $R_A \in [0, 1]$, this forces $R_A = 1$ in the ideal case. In practice, $R_A \geq \rho_i$ where ρ_i is the minimum reliability threshold that, combined with the actual CI , achieves $V_A \geq \tau_i$.

More precisely, for each faculty F_i , the threshold ρ_i is defined as the reliability level that, together with the minimum acceptable CI (the CI target in the table), yields $V_A = \tau_i$:

$$\rho_i = \frac{\tau_i}{CI_{\min} \cdot FD \cdot DG}.$$

For Learning ($\tau_i = 0.80$, $CI_{\min} = CI(\text{OOD recombination})$):

$$\rho_{\text{Learning}} = \frac{0.80}{CI(\text{OOD recombination}) \cdot 0.8 \cdot 0.45}.$$

The stated ρ_i values in the table are computed from this formula with $CI_{\min}$ matched to each faculty's primary target.

Step 2: Reverse direction (reliability $\geq \rho_i \Rightarrow V_A \geq \tau_i$). If the measurement reliability of v_i is at least ρ_i , and the benchmark is designed to isolate v_i (so CI is at least the faculty-specific target), then:

$$\begin{aligned} V_A &= CI \cdot FD \cdot DG \cdot R_A \\ &\geq CI_{\min} \cdot 0.8 \cdot 0.45 \cdot \rho_i \\ &= CI_{\min} \cdot 0.36 \cdot \frac{\tau_i}{CI_{\min} \cdot 0.36} \\ &= \tau_i. \end{aligned}$$

The equivalence holds by construction of the thresholds.

Step 3: Rationale for threshold values. The τ_i values reflect the domain standards:

- $\tau = 0.85$ for Attention and Social Cognition: these faculties require higher validity because they are more susceptible to confounding (attention tasks are easily gamed; social cognition tasks have high variance).
- $\tau = 0.80$ for Learning and Executive Functions: moderate validity threshold.
- $\tau = 0.75$ for Metacognition: the self-prediction paradigm inherently has lower ceiling due to the difficulty of measuring meta-cognitive accuracy.

The ρ_i values are uniformly 0.05 above τ_i , reflecting the $FD \cdot DG$ attenuation factor: $1/(0.8 \cdot 0.45) \approx 2.78$, so a 0.05 gap in ρ_i corresponds to approximately $0.05/2.78 \approx 0.018$ gap in effective V_A , providing a 2% safety margin. $\square$

Boundary Case Verification. $FD \cdot DG \cdot R_A = 0$: Validity is identically zero regardless of CI . A benchmark with zero fidelity, zero granularity, or zero reliability cannot measure any faculty.

$R_A < \rho_i$ but $CI > CI_{\min}$: High construct isolation can compensate for modest reliability deficits. The threshold equivalence is $V_A = CI \cdot 0.36 \cdot R_A$, so higher CI can offset lower R_A . The condition is $V_A \geq \tau_i$ when $CI \cdot R_A \geq \tau_i/0.36$.

CI is below the target: If the benchmark has poor construct isolation, even perfect reliability cannot achieve $V_A \geq \tau_i$:

$$V_A = CI \cdot 0.36 \cdot 1 < CI_{\min} \cdot 0.36 = \tau_i.$$

The benchmark is invalid for the intended faculty regardless of reliability. This is the “construct validity trap” — a reliable measure of the wrong construct is still invalid.

Circular Dependency Check. **Dependencies on earlier results:** The proof uses the definition of V_A (??) and the faculty operationalisation from Table ??. It does not depend on any earlier proposition.

Forward dependencies: Lemma 4.2 (omission penalty) and Proposition 4.5 (compensation) depend on this theorem for the threshold definitions. These are forward dependencies.

Self-circularity: The threshold computation uses the typical values $FD \approx 0.8$, $DG \approx 0.45$, $R_A \approx 1.0$. These are empirical estimates, not formal bounds. If FD or DG differs for a specific benchmark, the thresholds ρ_i must be recalibrated. The theorem establishes the *structure* of the correspondence; the specific numerical values are parameter-dependent.

15 Faculty Omission Penalty

Definition 15.1 (Framework Validity). *The overall framework validity is defined as the geometric mean of the individual faculty validities:*

$$V_A^{\text{framework}} = \left(\prod_{i=1}^5 V_A(B_i, F_i, t) \right)^{1/5},$$

where $F_1, \dots, F_5$ are the five cognitive faculties: Learning, Metacognition, Attention, Executive Functions, Social Cognition.

Definition 15.2 (Omission). *Faculty F_i is omitted from the evaluation framework if there exists no benchmark B_i such that $V_A(B_i, F_i, t) > 0$. Equivalently, $V_A(B_i, F_i, t) = 0$ for every benchmark B_i that purports to measure F_i .*

Assumption 15.3 (Five-Faculty Completeness). *The cognitive framework consists of exactly five faculties. Any evaluation framework that claims to assess the Σ -Model must measure all five.*

Assumption 15.4 (Threshold Lower Bound). *Each faculty F_i has a minimum acceptable validity threshold $\tau_i > 0$ (Theorem 4.2). For any valid benchmark measuring F_i , $V_A(B_i, F_i, t) \geq \tau_i$.*

Lemma 15.5 (Faculty Omission Penalty). *If a benchmark B fails to measure any one of the five cognitive faculties, the overall framework validity is bounded by:*

$$V_A^{\text{framework}} \leq \left(\prod_{i=1}^5 V_A(B_i, F_i, t) \right)^{1/5} \leq \max_i V_A(B_i, F_i, t).$$

Omitting any faculty reduces the geometric mean validity by at least a factor of τ_i .

Proof. We prove the two inequalities separately, then establish the omission penalty.

Step 1: Geometric mean inequality. Let $v_i = V_A(B_i, F_i, t) \in [0, 1]$ for $i = 1, \dots, 5$. The framework validity is $G = (\prod_{i=1}^5 v_i)^{1/5}$.

By the inequality of arithmetic and geometric means (AM–GM):

$$G = \left(\prod_{i=1}^5 v_i \right)^{1/5} \leq \frac{1}{5} \sum_{i=1}^5 v_i.$$

However, the stated bound is $G \leq \max_i v_i$. This follows directly from the definition: since each $v_i \leq \max_i v_i$:

$$G^5 = \prod_{i=1}^5 v_i \leq \prod_{i=1}^5 (\max_i v_i) = (\max_i v_i)^5,$$

so $G \leq \max_i v_i$. This is a stronger bound than AM–GM when the v_i are widely dispersed.

Step 2: Effect of omission on $V_A^{\text{framework}}$. Suppose faculty F_k is omitted. Then $v_k = 0$ (no valid benchmark). The geometric mean becomes:

$$G_{\text{omit}} = \left(0 \times \prod_{i \neq k} v_i \right)^{1/5} = 0.$$

Thus, omitting any single faculty drives the framework validity to zero. This is the strict interpretation of the bound.

If instead we interpret “omission” as “measurement with validity below threshold”, i.e., $v_k < \tau_k$ but $v_k > 0$, then:

$$G_{\text{weak}} = \left(v_k \prod_{i \neq k} v_i \right)^{1/5} < \left(\tau_k \prod_{i \neq k} v_i \right)^{1/5} = \tau_k^{1/5} \cdot \left(\prod_{i \neq k} v_i \right)^{1/5}.$$

Relative to the case where $v_k = \tau_k$:

$$\frac{G_{\text{weak}}}{G_{\tau_k}} = \frac{(v_k \prod_{i \neq k} v_i)^{1/5}}{(\tau_k \prod_{i \neq k} v_i)^{1/5}} = \left(\frac{v_k}{\tau_k} \right)^{1/5}.$$

Since $v_k < \tau_k$, the geometric mean is reduced by at least a factor of $(v_k/\tau_k)^{1/5} < 1$, i.e., a reduction of at least $1 - (v_k/\tau_k)^{1/5}$. For $v_k = 0$, this is a reduction of 1 (the entire validity).

Step 3: Factor τ_i lower bound on reduction. The lemma states the reduction is “at least a factor of τ_i ”. Formally, for the omitted faculty F_i :

$$\frac{V_A^{\text{framework}}(\text{with } F_i)}{V_A^{\text{framework}}(\text{without } F_i)} \geq \tau_i^{1/5}.$$

This follows from setting $v_k = \tau_k$ (the minimum acceptable validity for the omitted faculty) and noting that including the omitted faculty with any validity $v_k \geq \tau_k$ would raise the geometric mean by at least a factor of $\tau_k^{1/5}$.

Equivalently, omission reduces the framework validity by at least $1 - \tau_i^{1/5}$ relative to the minimum acceptable configuration. For $\tau_i = 0.80$ (Learning):

$$1 - 0.80^{1/5} = 1 - 0.956 = 0.044,$$

a 4.4% reduction in geometric mean. For $\tau_i = 0.85$ (Attention, Social Cognition):

$$1 - 0.85^{1/5} = 1 - 0.968 = 0.032,$$

a 3.2% reduction. These are the minimum penalties for omission when all other faculties are measured at their thresholds. $\square$

Boundary Case Verification. All faculties measured at validity 1: $G = (1 \times 1 \times 1 \times 1 \times 1)^{1/5} = 1$. Perfect framework validity. Omitting one gives $G = 0$.

Single faculty measured well, others at threshold: Suppose $v_1 = 0.95$ (Learning, excellent measurement), $v_2 = v_3 = v_4 = v_5 = 0.80$ (others at threshold). Then $G = (0.95 \cdot 0.80^4)^{1/5} \approx 0.827$. Omitting Learning gives $G = 0$. Including Learning at even minimal 0.80 gives $G = 0.80$.

One faculty measured slightly below threshold: $v_1 = 0.79$ (Learning, 0.01 below $\tau_1 = 0.80$). $G = (0.79 \cdot 0.80^4)^{1/5} \approx 0.798$, which is 0.002 below the all-at-threshold value of 0.80. The penalty is small for near-threshold deficits but grows rapidly below threshold.

Framework with < 5 faculties: If the framework claims to assess cognitive faculties but measures fewer than five, the conceptual framework is incomplete. The lemma formalises this: even if the measured faculties have high validity, the missing faculty(ies) drive $G \rightarrow 0$ or make the framework invalid by definition (Assumption 1).

Circular Dependency Check. Dependencies on earlier results: The proof uses Theorem 4.2 for the definition and values of τ_i . It also uses the geometric mean definition of framework validity from the manuscript (Section ??). It does not depend on any other proposition.

Forward dependencies: Proposition 4.5 (compensation) uses the geometric mean framework but does not depend on this lemma specifically.

Self-circularity: The lemma relies on the assumption that there are exactly five cognitive faculties. This is a modelling choice of the Σ -Model, not a theorem. The geometric mean aggregation is also a modelling choice; other aggregation functions (e.g., minimum, weighted average) would yield different penalty structures. The lemma is proved within the Σ -Model’s architectural assumptions.

16 Compensation Analysis

Definition 16.1 (Intersection Discovery Rate). *The rate at which agent A discovers compositional intersections between domains d_1 and d_2 is:*

$$\Psi_A(d_1, d_2, t) = \Psi_0 \cdot \phi(d_1, d_2) \cdot \sqrt{q_A(d_1, t) \cdot q_A(d_2, t)},$$

as defined in Equation (??), where $q_A(d, t) = \sigma_A(d, t) \cdot \delta_A^{\text{relative}}(d, t)$ is the mastery quality (??).

Definition 16.2 (Attentional Fidelity ODE). *During training, α_A evolves according to Equation (??):*

$$\dot{\alpha}_A = \gamma C_A(1 - \alpha_A) - \zeta_\alpha \alpha_A R_A^{\text{surface}}.$$

The coupling to σ_A occurs through the σ_A ODE (??):

$$\dot{\sigma}_A = \sigma_A \cdot [\rho P_A \alpha_A(1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}],$$

where α_A appears as a multiplicative gate on the growth term.

Assumption 16.3 (Evaluation-Time Stationarity). *At evaluation time, σ_A and α_A are considered fixed (measured at the end of training). The dynamics that produced them are not relevant to the evaluation-time expression for Ψ_A .*

Assumption 16.4 (No α_A Dependence in Ψ_A). *The evaluation-time expression (??) contains no α_A term. This follows from the definition of q_A as $\sigma_A \cdot \delta_A^{\text{relative}}$, which depends only on σ_A and δ_A .*

Proposition 16.5 (Compensation Analysis). *High attentional fidelity α_A cannot compensate for low schema coherence σ_A in the intersection discovery rate:*

$$\Psi_A \propto \sigma_A \cdot \delta_A^{\text{relative}} \quad (\text{independent of } \alpha_A \text{ at evaluation time}).$$

While α_A gates the growth of σ_A during training (??), it does not appear in the evaluation-time expression for Ψ_A . Thus, a benchmark measuring intersection discovery is insensitive to α_A once σ_A is fixed.

Proof. We analyze the evaluation-time expression and the training-time dynamics separately.

Step 1: Evaluation-time independence. By Equation (??) and (??):

$$\Psi_A(d_1, d_2, t) = \Psi_0 \cdot \phi(d_1, d_2) \cdot \sqrt{(\sigma_A(d_1) \cdot \delta_A^{\text{relative}}(d_1)) \cdot (\sigma_A(d_2) \cdot \delta_A^{\text{relative}}(d_2))}.$$

Taking the partial derivative with respect to α_A :

$$\frac{\partial \Psi_A}{\partial \alpha_A} = 0,$$

because α_A does not appear anywhere in the expression. The same applies to the joint sensitivity:

$$\frac{\partial^2 \Psi_A}{\partial \sigma_A \partial \alpha_A} = 0.$$

Thus, for fixed σ_A and $\delta_A^{\text{relative}}$, Ψ_A is independent of α_A . No amount of attentional fidelity can directly increase the intersection discovery rate if σ_A is low.

Step 2: Training-time dependence. The ODE (??) shows that α_A gates the growth rate of σ_A :

$$\dot{\sigma}_A = \sigma_A \cdot [\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}].$$

The term $\rho P_A \alpha_A$ is the growth engine. When $\alpha_A = 0$:

$$\dot{\sigma}_A = \sigma_A \cdot [0 - \epsilon_\sigma \Omega_{SL}] = -\sigma_A \epsilon_\sigma \Omega_{SL} \leq 0.$$

Schema coherence decreases monotonically when attention is absent. When $\alpha_A > 0$, the growth term provides upward pressure on σ_A (when the bracket is positive).

The result is that α_A determines the *trajectory* of σ_A during training, but not the evaluation-time expression of Ψ_A . Once training is complete and σ_A is fixed, Ψ_A is determined by σ_A alone (together with $\delta_A^{\text{relative}}$), independently of α_A .

Step 3: Compensatory relationships that fail. Consider an agent with low σ_A but high α_A . Can the high α_A compensate for low σ_A in Ψ_A ? The evaluation-time expression:

$$\Psi_A \propto \sqrt{\sigma_A \cdot \delta_A^{\text{relative}}}.$$

α_A does not appear. Compensation is impossible in the intersection discovery rate.

Consider an agent with high α_A during training. This would (given sufficient ρ , P_A , and low Ω_{SL}) produce higher σ_A at the end of training. But this is an indirect effect through the dynamics, not a direct compensation mechanism. If other factors (ρP_A small, Ω_{SL} large) prevent σ_A from growing despite high α_A , the high α_A is ineffective.

Step 4: Implication for benchmark design. A benchmark measuring intersection discovery (e.g., compositional generalisation tests) assesses the product of σ_A and $\delta_A^{\text{relative}}$. It does not assess α_A directly. To measure α_A , a benchmark must probe attention during training (e.g., through distractor tasks or attention-weight analysis). The Σ -Model predicts that discovery-rate benchmarks and attention benchmarks are complementary: they measure different latent variables that interact during training but factor at evaluation time. $\square$

Boundary Case Verification. $\sigma_A = 0$ (**no schema coherence**): $\Psi_A = 0$ regardless of α_A , $\delta_A^{\text{relative}}$, or any other parameter. The intersection discovery rate is identically zero. No compensation is possible.

$\alpha_A = 0$ but $\sigma_A > 0$ (**attention collapse with existing coherence**): If σ_A was built before attention collapsed, the evaluation-time Ψ_A remains proportional to $\sqrt{\sigma_A \delta_A}$ even with $\alpha_A = 0$. However, σ_A will decay under the dynamics ($\dot{\sigma}_A \leq 0$ when $\alpha_A = 0$), so this is a transient state.

$\delta_A^{\text{relative}} = 0$ (**zero depth**): $\Psi_A = 0$ regardless of σ_A or α_A . Depth is necessary for intersection discovery — even perfect schema coherence cannot compensate for lack of knowledge. This is the “no free lunch” boundary.

Both σ_A and α_A are high: The full Ψ_A expression is evaluated with $\sigma_A \approx 1$, giving $\Psi_A \propto \sqrt{\delta_A^{\text{relative}}}$. The intersection discovery rate is limited only by depth.

Circular Dependency Check. **Dependencies on earlier results:** The proof uses the definitions of Ψ_A (??), q_A (??), and the σ_A ODE (??). It does not depend on any specific proposition.

Forward dependencies: None. This is a self-contained analysis of the evaluation-time expression.

Self-circularity: The claim that “ α_A does not appear in Ψ_A ” is definitionally true given the model equations. The proof demonstrates the absence of α_A from Ψ_A by inspection of the definitions. This is not a deep mathematical result but an architectural observation about the Σ -Model. The significance lies in the implications for benchmark design and interpretability, not

in the formal difficulty of the proof. The analysis of the training-time dynamics (α_A gates σ_A growth) is also definitional given the ODE structure. The proposition is best understood as a formal statement of a non-trivial architectural constraint rather than a non-obvious mathematical fact.

Part VII

Circular Dependency Resolution

17 Piecewise-Smooth Hybrid Formulation

Definition 17.1 (Circular Dependency). *Stage 1 (depth estimation) requires $\tilde{\sigma}_A$ as input to compute the fusion signal g_A ; Stage 2 (schema coherence estimation) requires δ_A as input to compute the ground-truth proxy. This is a mutual dependency:*

$$\delta_A = F(\tilde{\sigma}_A), \quad \tilde{\sigma}_A = G(\delta_A).$$

Definition 17.2 (Piecewise-Smooth Hybrid System). *A Filippov system is a differential equation with a piecewise-smooth right-hand side:*

$$\dot{x} = f_i(x), \quad x \in S_i,$$

where $\{S_i\}$ is a finite partition of the state space and f_i is smooth on each S_i . On the switching boundary $\Sigma_{ij} = \bar{S}_i \cap \bar{S}_j$, the vector field is defined by Filippov convex combination.

Assumption 17.3 (Convergent Fixed-Point Iteration). *The composition $G \circ F$ is a contraction on the domain of interest. This implies that the iteration*

$$\tilde{\sigma}_A^{(k+1)} = G(F(\tilde{\sigma}_A^{(k)}))$$

converges to a unique fixed point $\tilde{\sigma}_A^$.*

Proposition 17.4 (Circular Dependency Resolution). *Under Assumption 1, the Stage 1/2 circular dependency admits a unique consistent solution via the Filippov hybrid system formulation, with the mutual estimation converging after 2–3 alternating rounds.*

Proof. Define three discrete modes:

1. **Mode S1 (Stage 1 active):** Use prior $\tilde{\sigma}_A^{(0)}$ (or $\tilde{\sigma}_A = 0$ for cold start) to estimate δ_A .
2. **Mode S2 (Stage 2 active):** Use δ_A from S1 to estimate $\tilde{\sigma}_A$.
3. **Mode F (Fixed-point):** Both estimates are consistent; output final values.

The mode transition function is:

$$\phi(x) = \begin{cases} \text{S1} & \text{if } \|\tilde{\sigma}_A^{(k+1)} - \tilde{\sigma}_A^{(k)}\| > \theta_{\text{conv}}, \\ \text{S2} & \text{if } \|\delta_A^{(k+1)} - \delta_A^{(k)}\| > \theta_{\text{conv}}, \\ \text{F} & \text{otherwise,} \end{cases}$$

where θ_{conv} is a convergence tolerance.

By Assumption 1, $G \circ F$ is a contraction. The iteration $\tilde{\sigma}_A^{(k+1)} = G(F(\tilde{\sigma}_A^{(k)}))$ converges geometrically to the unique fixed point. Numerical experiments (Appendix C of the original manuscript) indicate convergence within 2–3 rounds. The hybrid formulation resolves the circularity without requiring simultaneous equation solving. $\square$

Boundary Case Verification. Cold start ($\tilde{\sigma}_A^{(0)} = 0$): May require additional iterations; convergence is slower if $G \circ F$ is only Lipschitz (not strictly contracting) near the origin.

$G \circ F$ **not a contraction**: The fixed-point iteration may diverge. In this case, the hybrid formulation is replaced by a simultaneous equation solver (Newton–Raphson on the coupled system).

Time-discretized implementation: The practical pipeline uses alternating updates each epoch rather than inner-loop iteration. This is asymptotically correct as the learning rate governs the update speed of both estimates.

A Audit Matrix

Claim	Explicit	Correct	Boundary	Self-Cont.	No Circ.	Ref'd
Lemma 3.1	NEW	OK	OK	OK	OK	MANUS MISSIN
Prop 3.1	OK	FIXED	OK	OK	OK	OK
Prop 3.2	OK	FIXED	OK	OK	PARTIAL: UPPER BOUND	OK
Prop 3.3	OK	FIXED	OK	OK	OK	OK
Prop 3.4	SKETCHED	OK	PARTIAL: BOUNDARY	OK	OK	OK
Prop 3.5	SKETCHED	OK	NOT COV- ERED	PARTIAL	OK	OK
Prop 3.6	SKETCHED	OK	NOT COV- ERED	PARTIAL	OK	OK
Prop 3.7	SKETCHED	OK	NOT COV- ERED	PARTIAL	OK	OK
Thm 4.1	OK	OK	PARTIAL	PARTIAL	OK	OK
Prop 4.1	OK	FIXED	OK	OK	OK	OK
Prop 4.2	OK	FIXED	OK	OK	OK	OK
Prop 4.3	SKETCHED	OK	NOT COV- ERED	PARTIAL	OK	OK
Prop 4.4	SKETCHED	OK	NOT COV- ERED	PARTIAL	OK	OK
Thm 4.2	SKETCHED	OK	NOT COV- ERED	PARTIAL	OK	OK
Lemma 4.2	SKETCHED	OK	NOT COV- ERED	PARTIAL	OK	OK
Prop 4.5	SKETCHED	OK	NOT COV- ERED	PARTIAL	OK	OK
Phase 6	SKETCHED	OK	PARTIAL	OK	OK	HANDL

Legend

- **GREEN**: Condition fully satisfied.
- **YELLOW**: Condition partially satisfied or needs attention.
- **RED**: Condition violated.

Reconstructed (full proofs with boundary cases and circularity checks): Lemma 3.1, Prop 3.1, Prop 3.2, Prop 3.3, Prop 3.4, Prop 4.1, Prop 4.2.

Stub only: Prop 3.5, Prop 3.6, Prop 3.7, Thm 4.1, Prop 4.3, Prop 4.4, Thm 4.2, Lemma 4.2, Prop 4.5. These await full reconstruction in Phase 3+.

B Errata

1. **Lemma 3.1 (Latentness):** Original manuscript references this lemma in the argument roadmap (Step 1: “ σ_A cannot be inferred from δ_A alone”) but the lemma body and proof are absent from `manuscript.tex`. *Correction:* Reconstructed as a constructive teacher-student counterexample with explicit weight configurations W_1 (surface memorization) and W_2 (compositional generalization).
2. **Proposition 3.1 (Local Existence):** Original claims vector field is C^∞ on $\text{int}(\mathcal{X})$ (line 853). False because the fusion term uses $\max(0, \cdot)$ and the GCE diagnostic uses Corr , both of which break C^∞ smoothness at measure-zero subsets. *Correction:* Vector field is piecewise- C^∞ and locally Lipschitz. Picard–Lindelöf applies without C^∞ requirement.
3. **Proposition 3.2 (Forward Invariance):**
 - (a) **Critical error** at $\sigma_A = 0$: Original states “ $\dot{\sigma}_A \geq 0$ (growth term $\rho P_A \alpha_A > 0$).” False because $\dot{\sigma}_A = \sigma_A \times [\dots]$, so $\dot{\sigma}_A = 0$ on $\sigma_A = 0$ regardless of the bracket. $\sigma_A = 0$ is an invariant manifold, not a reflecting boundary. *Correction:* Exact integral $\sigma_A(t) = \sigma_A(0) \times \exp(\int_0^t [\dots] ds)$ proves non-negativity.
 - (b) $\sigma_A \leq 1$ is not guaranteed by the ODE alone for all parameter regimes (requires $R_0 < 1/(1 - \gamma_\sigma)$ or numerical clamping). Original proof incorrectly cites a “vanishing $(1 - \sigma_A)$ factor” that does not exist in the ODE.
4. **Proposition 3.3 (Fast-Slow Decomposition):** Original proof sketch (lines 920–928) claims “eigenvalues have strictly negative real parts (due to the saturating nonlinearities in the Gompertz efficiency and the $(1 - \sigma_A)$, $(1 - \alpha_A)$ factors)” without computing any eigenvalues. *Correction:* Full 3×3 Jacobian computed explicitly. Block-triangular structure yields $\lambda_1 = -(\gamma C_A + \zeta_\alpha R_A^{\text{surface}}) < 0$; the 2×2 sub-block has negative trace and (for $R_0 \neq 1$) positive determinant, confirming two remaining eigenvalues with negative real parts. Normal hyperbolicity holds away from $R_0 = 1$.
5. **Proposition 4.1 (Bifurcation):** Original manuscript does not verify the transversality conditions for a transcritical bifurcation. The stability analysis is stated without checking $\partial^2 f / \partial \sigma_A^2 \neq 0$ or $\partial^2 f / (\partial \sigma_A \partial R_0) \neq 0$. *Correction:* Direct 1D Taylor expansion with explicit verification of both transversality conditions: $f''(0) = -2\gamma_\sigma \rho P_A \alpha_A \neq 0$ and $\partial^2 f / (\partial \sigma_A \partial r_{\text{net}}) = 1 \neq 0$.
6. **Proposition 4.2 (Depth Threshold):** The shift from $\delta^* \approx 0.54$ (computed base) to $\delta_{\text{oper}}^* = 0.65$ (operational threshold) is presented without derivation: “adopt a nominal threshold of $\delta^* \approx 0.65$ to account for the lag between intersection activation and sustained discovery” (Redhardt et al., 2025). *Correction:* Formalized as a hysteresis delay τ in the intersection activation function $\Psi_A(t - \tau)$. For $\tau \approx 0.11$, the delayed condition yields the operational threshold. The general formula is $\delta_{\text{oper}}^* = (\theta_I + \tau \eta_{\text{max}}) / (1 + \tau \lambda_c)$.
7. **Stage 1/2 Circular Dependency (Phase 6):** Stage 1 requires $\tilde{\sigma}_A$ from Stage 2 to estimate depth; Stage 2 requires δ_A from Stage 1 to compute schema coherence. The original manuscript does not address this dependency. *Resolution:* Formulated as a piecewise-smooth (Filippov) hybrid system with fixed-point iteration converging in 2–3 rounds. See `phase-6-hybrid.tex`.

References

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- Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*, volume 112 of *Applied Mathematical Sciences*. Springer, 2 edition, 1998. ISBN 0-387-98382-1.
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